INTRODUCTORY CHEMICAL THERMODYNAMICS

Chuck Pearson
Tusculum University

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CHAPTER OVERVIEW

1: Measurement, energy, and the quantites of state

This introduction is taken from my first introduction of the content to students in CHEM 411 at Tusculum University in Fall 2019 - the first to study from this material. I still am trying to build this material with these principles in mind.

You have had enough courses that ask you to remember a ton of facts and terms. And you will still need to make sure you know old and new terms for this course, and you will need to be certain you are using those terms precisely, but there's no good means to test you on the terms themselves as a means to make them meaningful.

You've also had enough experience being given equations that are rules and having to use those equations (frequently without much context) to solve simple one- or two-step problems. Those skills are important to build, and I will check skills on equation sets from time to time here, especially when those equations involve calculus. But they don't need to be our primary business either.

The development of physical chemistry, especially chemical thermodynamics, is a necessarily theoretical discipline. Energy is a constructed physical quantity we use to make our language understandable across disciplines - it doesn't have the tangibility of force or the broad common meaning of temperature. We can take some of that language and directly apply it to behavior we do see - but even then, much of that is the behavior of gases, which are atoms and molecules spread out over wide distances from one another. We have to make some effort to describe that behavior meaningfully. That effort to describe that behavior demands the use of our imaginations, and demands development of a new language, and demands mathematical descriptions to fit that language - *and all must fit together to make a coherent whole.*

Our primary business in chemical thermodynamics is the development of a good theory of the transfer of energy through physical and chemical change. My writing in this text, and the problems I assign for you to solve, will all be in support of developing this theory in a meaningful way, in a way that allows you both to understand the richness of the fundamental ideas of energy and to describe energy transfer in practical contexts.

There will be *both* English-language concepts *and* mathematical concepts that we will use *precisely*. I can't emphasize enough how important it is to not be careless with how you use the mathematics you'll be developing in this course - tricks you've employed in the past are much less likely to work, and points where you've taken mathematical shortcuts in the past are much more likely to lead to dead ends now.

But our process in this course is going to be very intentional as well. I will do my best to not assume that you automatically understand some of the core math concepts, particularly calculus concepts. For many - perhaps most - of you, this is the first serious physical science problem solving you will ever do that involves calculus. My development of methods for this kind of problem solving will be patient and total.

I joke that I'm interested in making good theoreticians in this course. But you've had a host of good experience in lab sciences where your work is very practical. I want you to master the symbolic and conceptual logic necessary to develop physical ideas using the most powerful scientific tools: pencil, paper, and your mind.

1.1: The language of measurement in physical chemistry

In order to be good theoreticians, we need tools of our theory. And because this is a physical science, those tools are primarily those of physical measurement. The building blocks of those tools are the same as those that you've encountered in physics and chemistry courses before now: the base SI (*Système international*) units.

Physical quantity	**Symbol**	**SI quantity (*mks*)**	**Abbreviation**
Length	x (or Δx, or L)	meters	m
Mass	m	kilograms	kg
Time	t (or Δt)	seconds	s
Electric current	I (or J)	Amperes*	A*
Temperature	T	Kelvin	K
Count of substance	n	moles	mol

Table 1: The base SI units, associated with the physical quantities that they measure. Format of table taken from Monk's Physical Chemistry.

**Depending on the system, either the ampere or the coulomb can be used as the base unit associated with electricity; many definitions are easier to build with the ampere than the coulomb, however. Remember that 1 C = (1 A)•(1 s).*

You have studied all of these concepts previously. This is likely the first time you've seen all the base SI units assembled in one place, however. In this course, literally all the units we will build are going to be compounds of these units. Some of these compound units will be very straightforward; the units of area and volume, m^2 and m^3, are compounds of a single unit, the length. Others are going to be more sophisticated.

To build these compound units, let's remind ourselves of some key ideas from physics.

Force is one of the first compound units in physics that we get to know well. The SI unit for force is the **newton**. It is built on the back of the definition of force, which is that interaction necessary to change the velocity of a mass, the definition that's commonly known as Newton's second law:

$$F = ma(\text{single force acting on single object})\text{Units: } 1\ \mathrm{N} = (1\ \mathrm{kg})(1\ \mathrm{m\ s}^{-2})$$

The symbol a, from classical mechanics, represents acceleration, which is a compound unit in and of itself. For very deliberate reasons I don't want to consider acceleration too deeply mathematically right now; if you remember that *acceleration* is synonymous with "rate of change of velocity", you have what you need to have right now.

The newton is a compound of all three classic base SI units: mass, length, and time. It forms the foundation for other ideas as well, including the idea of **work**. We will explore many increasingly precise formal definitions of work as we move forward in the course, but the fundamental nature of work is applying a force to an object as that object moves. If that force has a component that is parallel to the motion, the work done is positive; if that force has a component that is antiparallel to the motion, the work is negative.

$$w = F\Delta x\cos\theta(\text{single force acting on single object})\text{Units: } 1\ \mathrm{J} = (1\ \mathrm{N})(1\ \mathrm{m})$$

(Work may have been symbolized in courses you have taken previously with a capital block or cursive W. In physical chemistry, work is always symbolized with a lower-case w, and as we will find later in the course, the lower-case symbol means as much as the letter.)

There is one other key relationship that comes out of classical mechanics. When work is applied via a constant force to accelerate an object, there is an expression that becomes quite useful - an expression for the mass of that object multiplied by its velocity squared. When framed to fit, we can easily prove that work is equal to the change in this quantity, which we have historically defined as kinetic energy:

$$K = \frac{1}{2}mv^2\text{Units: } 1\ \mathrm{J} = (1\ \mathrm{kg})(1\ \mathrm{m}^2\mathrm{s}^{-2})$$

The most important thing to recognize about these two definitions is the units. They're exactly identical. We use the name of the man who equated mechanical energy with heat, James Joule, to connect these two ideas; the SI unit of energy is a compound unit, the **joule**, equivalent to one kilogram times the square of one meter per second, or one newton times one meter.

There are a lot of physical measurements that we can come up with that are specific to one thing or another, and that only have a single application. The joule is a measure that we will apply to a host of phenomena that will connect back in one way or another to energy. That in turn provides us with one of the most important properties of the energy concept; its *holistic* nature.

We can't apply force to much of anything outside of specific pushes and pulls; force is not an idea that has much application in chemistry. We will rarely, in classical mechanics, have any occasion to count up enough of anything to constitute the mole; most applications of the mole concept are in chemistry. There are other specific measures (drag, or extinction coefficient, or genetic distance) that apply only in the context of a narrow set of problems in a narrowly defined discipline.

Energy is the one major concept that can describe phenomena in physics, in chemistry, in biology, in engineering, and beyond. All disciplines need to explain work done; all disciplines need to explain heat transfer; all disciplines need to explain stored energy. In studying energy in physical chemistry, we are studying not merely physics or chemistry but principles applicable across the basic and applied sciences.

1.2: The meaning and measurement of temperature

One of the key ways that we will need to describe energy is in direct connection to the temperature of a system. One of the foundational ideas of *statistical mechanics* is a model for the behavior of gas molecules called the *kinetic theory of gases* - it's a model built on considering the particles in a gas to be very small, noninteracting, and colliding in perfectly elastic ways. (The Wikipedia article on the kinetic theory is a wonderful primer with a very understandable derivation.) The kinetic theory also provides our first definition of temperature, which is *a measure of the average kinetic energy of the particles of a gas at a state of thermal equilibrium.* Mathematically, that measure is

$$T = \frac{2K_{tot}}{3Nk_B} \text{Units: K}$$

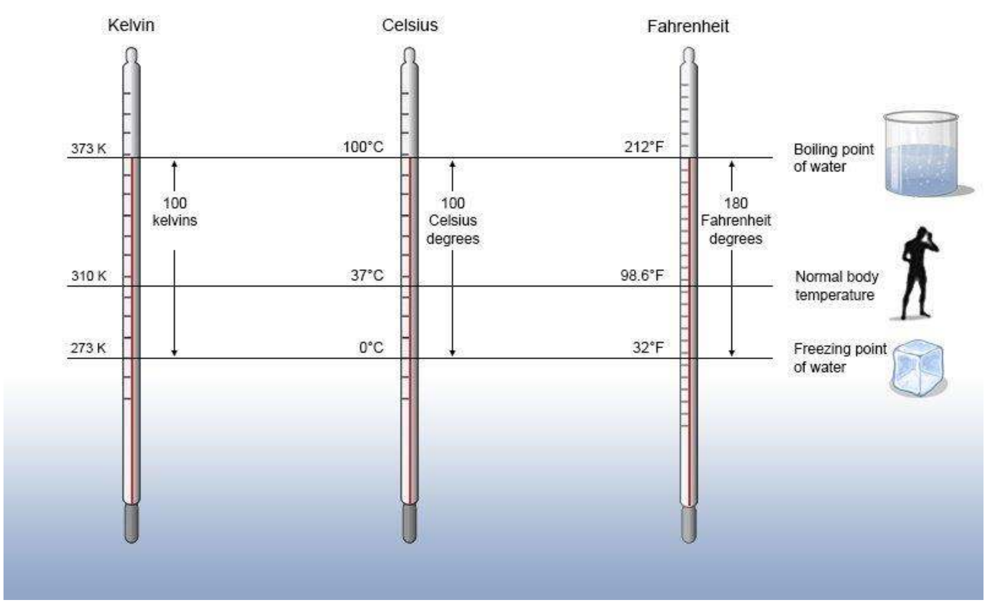
We can prove this result with the kinetic theory, but we will take it as a definition for now. The average kinetic energy is hidden in the expression of total kinetic energy divided by N, the count of molecules present.

This forces the units of the genuinely new quantity k_B to be Joules per Kelvin. This is a constant, our first of the course; it is known as the *Boltzmann constant*, after the German thermodynamics scholar Ludwig Boltzmann who first described it:

$$k_B = 1.38065 \times 10^{-23} \text{ J K}^{-1}$$

We will explore the meaning of this constant and its connection to other ideas in thermodynamics as we move forward in the course. Multiplying k_B by temperature gives units of energy, and it is not uncommon when talking about the energy of individual molecules to describe their energy in units of k_BT.

The direct proportionality between average kinetic energy of molecules and temperature also enforces a system of measuring temperature where the temperature is zero when the average kinetic energy of all molecules is zero - a circumstance whose existence is only theoretical, never actually observed. It also enforces the impossibility of negative temperature. We call the Kelvin scale an *absolute* scale for that reason - every temperature is positive and has a neat proportional connection to properties of that gas. No other temperature scale has such a neat proportional connection, which is why - even though the Celsius scale and the Fahrenheit scale are convenient to use for other reasons, we will overwhelmingly prefer the Kelvin scale when doing thermodynamic calculations.



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It's worth a reminder of how the Celsius and Fahrenheit scales are structured compared to the Kelvin scale. Celsius was deliberately scaled to have 100 degrees between the freezing and boiling points of water; the Celsius scale was set prior to the Kelvin scale, and the value of the Boltzmann constant turns out to be dependent upon an absolute temperature scale where there are 100 units of temperature between the freezing and boiling point of water. The development of the Fahrenheit scale is more obscure, but the original intent was to set temperatures so that the normal body temperature is 100 degrees. It also turns out to be more convenient to define Fahrenheit based on freezing and boiling points of water, however.

1.3: The variables of state, pressure units, and the ideal gas law

Temperature is only one of the key ways that we can describe the properties of a gas. Counting up the number of molecules - or, more reasonably, the number of *moles* - is another.

These are two of the four conventional *variables of state* (or *state variable*) for a gas. The variables of state are considered the ways of describing the *entire* condition of an ideal gas.

State variable	Symbol	SI Unit (mks)	Abbreviation	Base SI units
Temperature	T	Kelvin	K	K
Count of substance	n	moles	mol	mol
Volume	V	cubic meters	m^3	m^3
Pressure	P	Pascals	Pa	kg m^{-1} s^{-2} (or N m^{-2})

Volume units are very straightforward; they're length units in three dimensions, in keeping with a volume being proportional to any length dimension of an object cubed. Of course, the base SI units for volume, cubic *meters*, are *very* large indeed; nobody would actually want to lift a cube of anything that was a meter of length on each side. This is why most small-scale exercises will set units of volume to cubic *centi*meters, or the size of a cube 1 centimeter on each side; this is much more manageable.

Many modern physical chemistry textbooks will make mention of a cubic *deci*meter, the size of a cube 1 decimeter (or 10 centimeters) on each side. 1 dm^3 is just another way to describe one *liter*, which is the common metric unit of volume. Because of the cubing of 10, there are 1000 liters in a cubic meter; there are 1000 cubic centimeters in a liter, which is why the cubic centimeter and the milliliter are frequently equated with one another.

$$1\ \mathrm{L} = 1\ \mathrm{dm}^3 = 1000\ \mathrm{cm}^3 = 1000\ \mathrm{mL} 1\ \mathrm{m}^3 = 1000\ \mathrm{dm}^3 = 1000\ \mathrm{L}$$

Pressure units are much more plentiful, and much more convoluted. *Pressure* is defined simply as a force per unit area. The base SI unit of pressure is the *Pascal*, which is equivalent to SI measure of force (the Newton) divided by the SI measure of surface area (the square meter). This is a *very* small unit of pressure, approximately one-one hundred thousandth normal atmospheric pressure.

Historically standard atmospheric pressure has been defined by the height of a column of mercury that is supported by that pressure, in the nature of an experiment designed by one of Galileo Galiei's apprentices, Evangelista Torricelli. In his honor, a height of 1 millimeter of mercury in a barometric column is called a *torr*; standard atmospheric pressure, also known as *one atmosphere*, is equivalent to a height of 760 mm of mercury in this kind of column; this turns out to be about 101,300 Pascals (or 101.3 kiloPascals).

Most chemists use the atmosphere as their reference atmospheric pressure, but some prefer a truly metric unit, the *bar*, which is set to exactly 100,000 Pascals (exactly 100 kPa). Meterological pressure is frequently communicated in *millibars*.

$$1\ \mathrm{atm} = 760\ \mathrm{torr} = 1.013\ \mathrm{bar} = 101\,300\ \mathrm{Pa} 1\ \mathrm{bar} = 750.2\ \mathrm{torr} = 0.9872\ \mathrm{atm} = 100\,000\ \mathrm{Pa}$$

For reasons that are purely historic, we will prefer the atmosphere as our units of pressure - but there will be times when any of the four will be preferable. Vapor pressures are far more frequently communicated in torr (which, again, is the same thing as mm Hg). When we do computations in base-SI units, we will need Pascals; and again, bars are metric derivations from the Pascal. Atmospheres are simply most frequently used and most easily explained - a multiple of normal atmospheric pressure.

If we take the kinetic theory's two conditions of *ideality* of gases - the essentially zero volume of the gas molecules, and the lack of attraction or repulsion between gas molecules - as valid, we find that the ratio of pressure times volume of an ideal gas to number of moles times temperature is a *constant*:

$$R = \frac{PV}{nT} R = 0.082058 \mathrm{L\ atm\ mol^{-1} K^{-1}} = 8.31447 \mathrm{J\ mol^{-1} K^{-1}}$$

This is the equation more commonly written as $PV = nRT$, the ideal gas law. R, like the value of k_B, is a constant that will turn up every now and again in our computations. Like the definition of temperature, we *can* derive this law from first physical principles; however, this is a familiar enough statement to us that we will simply take it as something we know, as a straightforward starting point.

There's one additional thing to note about these two constants we've worked with, the value of k_B and the value of R. You see a hint of that thing in the exponent on the value of k_B, the exponent of 10^{-23}. Divide the value of R by the value of k_B and make a note of what comes out:

$$\frac{R}{k_B}=\frac{8.31447\mathrm{J\ mol^{-1}K^{-1}}}{1.38065\times 10^{-23}\mathrm{J\ K^{-1}}}=6.0221\times 10^{23}\mathrm{mol^{-1}}\rightarrow\frac{R}{k_B}=N_A$$

R and k_B are, in many ways, the exact same constant. R, on the one hand, deals with an energy per *mole* per Kelvin; k_B deals with an energy *per molecule* per Kelvin. The ratio of R to k_B is the familiar value of Avogadro's number; this connection can prove useful for us as we move forward.

1.4: The ideal gas law, functions and derivatives

The ideal gas law gets taught and learned in a easily memorized way:

$$PV = nRT$$

However, when it comes to visualizing the realities of the ideal gas law in graphical form, it's more useful to rearrange the equation. Most presentations of ideal gas behavior as a function of a variable of state make pressure the dependent variable:

$$P(n,T,V) = \frac{nRT}{V}$$

Note that the left side is not merely pressure *P*, but is pressure as a function of the other three state variables - $P(n, T, V)$. Because this is a function of more than one variable, it resists most of the understanding of calculus we obtain early in our mathematical education, where we only take derivatives of functions of a single variable. Taking derivatives of one variable when other variables are changing isn't often useful for explanation.

So we say this expression has no meaning:

$$\frac{dP}{dT} \rightarrow \text{meaningless}$$

However, let's say we're only interested in the rate of change of pressure as temperature changes, leaving number of moles constant and volume constant (we describe this as change behavior in a closed, rigid container). That *does* have meaning, but we need to express it in a new way, a way we haven't seen before. We describe this as a *partial derivative*, in this case a partial derivative of pressure with respect to temperature, with moles and volume constant. We notate this derivative like this:

$$\left(\frac{\partial P}{\partial T}\right)_{n,V}$$

Using the lower-case Greek letter ∂ is our indicator that this is a *partial* derivative that will focus on only one variable; the variables to be held constant (n, V) are placed in the subscript at lower right. Therefore the partial derivative is equivalent to a derivative of a single variable with all the constants factored out:

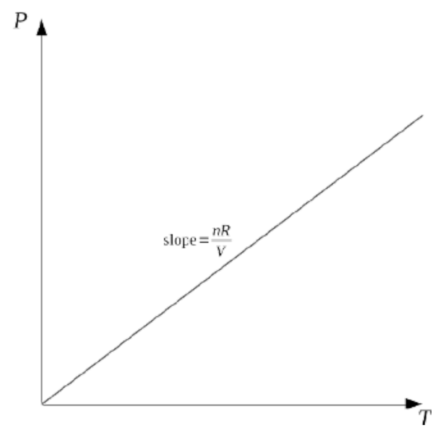
$$\left(\frac{\partial P}{\partial T}\right)_{n,V} = \left(\frac{\partial}{\partial T}\frac{nRT}{V}\right)_{n,V} = \frac{nR}{V}\frac{d}{dT}T$$

But the first derivative with respect to T of T is just 1, and this expression can be easily simplified:

$$\left(\frac{\partial P}{\partial T}\right)_{n,V} = \frac{nR}{V}$$

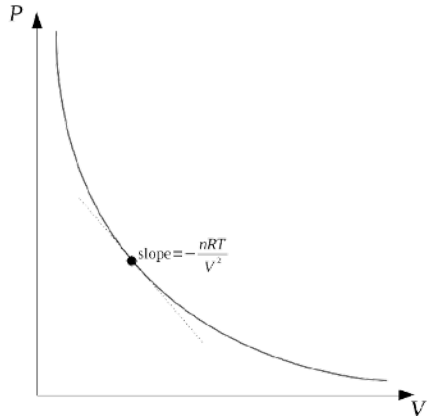
This means that if moles and volume are held constant, the derivative of pressure with respect to temperature must be a constant as well.

This is not surprising!



The relationship between pressure and temperature is a linear relationship, after all; we could just solve the ideal gas law for P/T and we'd get the same slope nR/V. A linear relationship between a dependent and an independent variable is a relationship where the derivative of the dependent variable doesn't change, because the slope of the graph isn't changing.

There are many relationships between the variables of state that turn out to be linear in this way. Not all of them do, however. The most important example of a nonlinear relationship between variables of state is the relationship between pressure and volume:



Just a quick scan of this graph indicates that this *won't* give us a constant slope at every position the way the pressure vs. temperature graph did. And indeed, we run into an additional application of the calculus when we go to take the derivative of a function of pressure with respect to volume, holding temperature and moles constant - the volume is in the denominator, and derivatives of negative powers of a quantity must be negative themselves:

$$\left(\frac{\partial P}{\partial V}\right)_{n,T} = \left(\frac{\partial}{\partial V}\frac{nRT}{V}\right)_{n,T} = nRT\left(\frac{d}{dV}\frac{1}{V}\right) = nRT\left(-\frac{1}{V^2}\right)$$

$$\left(\frac{\partial P}{\partial V}\right)_{n,T} = \left(-\frac{nRT}{V^2}\right)$$

This is a little more involved, but again it's not surprising. At every point, as pressure is increased, volume of the container decreases; the rate of change of pressure with respect to volume is negative. The rate of change is steepest at small values of V and shallowest at large values of V; that reflects the term of V^2 in the denominator of the derivative. The mathematics of the derivative predicts the trend of the graph.

We take derivatives in physical chemistry with this purpose in mind. We are using these derivatives to build our explanatory power for the trends that are fundamental to the relationship between the variables of state.

1.5: Ideal vs. real gases and the van der Waals equation

Consider a simple application of the ideal gas law. Let's take an oxygen tank of rigid walls, with a fixed volume of 40.0 L. Let's maintain this tank at room temperature of 298 K. Let's fill this gas container with 1.000 kg of oxygen gas. We can show that this mass of oxygen corresponds to 31.25 moles of gas (prove this to yourself!). We can then use that number of moles, the volume of the tank, and the temperature to find the pressure the gas is under:

$$PV = nRT \rightarrow P = \frac{nRT}{V} = \frac{(31.25\text{mol})(0.082058\text{L atm mol}^{-1}\text{K}^{-1})(298\text{K})}{40.0\text{L}} = \mathbf{19.1\ atm}$$

Now think about the ideal gas law computation we've just completed. There's nothing in that computation that's dependent on the gas being oxygen at all; the only thing in the setup of that equation that did depend on the identity of the gas was determining that there was 31.25 mol of gas in 1.000 kg of oxygen. 31.25 moles of *any* gas, according to the ideal gas law, would apply a pressure of 19.1 atm on the walls of its container.

The ideal gas law, of course, is one of the great general chemistry fibs. It doesn't work in real life.

Now, it comes *close* to working in real life under the conditions we care about the most; relatively warm temperatures and relatively low pressures, conditions where the space between molecules is great and the molecules can zip past one another without substantial attraction or repulsion. We teach the ideal gas law not because it's a perfect description of how gases behave at all conditions, but because it comes acceptably close to describing how gases behave at the conditions we're most likely to experience, and it gives a passable first estimate even outside of those conditions. The ideal gas law clearly communicates that putting 1.000 kg of an elemental gas into a 40.0 L container is going to result in a *very* high pressure. Even if it's not a perfect computation, it's useful.

One of our first big theoretical hurdles is this: when we have conditions where the ideality of the gases isn't something we can assume, when the molecules are big and when the molecules attract one another substantially, how do we handle the correction?

There are two approaches we can take. We'll take the more rational approach first.

Think about the quantities of state. The number of molecules present - the count of moles - doesn't change when the attraction or the size of molecules changes. Neither does the kinetic energy - the assumptions of ideality of a gas have to do with the *size* of molecules, not the *mass*. So we can leave *n* and *T* alone when we correct for nonideality.

The size of the molecule obviously impacts *volume*. *V* in the ideal gas law works best, it turns out, when it represents the amount of space in the gas *not taken up by gas molecules*. When we go from molecules that are infinitely small to molecules that have real size, the volume of space available is going to be thrown off by that size, and thrown off more dramatically as the gas molecules get bigger. So we have to correct by subtracting out the amount of space taken up by the molecules, which we do this way:

$$V_{corr} = V - nb b\text{units: L mol}^{-1}$$

The correction we've introduced is a quantity called the *molar volume of gas molecules*, or simply the *van der Waals b constant*, after the Dutch thinker who developed this correction. The larger the size of the gas molecule, the larger *b* is. It's as straightforward as can be.

It also stands to reason that attraction or repulsion between molecules impacts pressure; the more gas molecules attract, the less the pressure is (because the attraction draws the molecules together and relieves the pressure), and the more gas molecules repel, the greater the pressure is (because the repulsion drives the molecules apart and pushes against the walls of the container). A corrected pressure that would fit the ideal gas law would add in a term that accounts for the magnitude of the attractive force between molecules. Unfortunately, developing a mathematical relationship for this term is not as straightforward as correcting the volume, because of the complex nature of that attraction even under conditions that remain close to ideality. The relationship that van der Waals developed to establish the value of his *van der Waals a constant* (which doesn't have a secondary name) is simply going to be described here without proof:

$$P_{corr} = P + \frac{n^2 a}{V^2} a\text{units: L}^2\text{atm mol}^{-2}$$

Therefore a full correction to the ideal gas law, which we call the *van der Waals equation*, keeps the *nRT* side intact but changes the left side substantially:

$$\left(P+\frac{n^2a}{V^2}\right)(V-nb)=nRT\text{van der Waals equation}$$

The values of *a* and *b* are unique based on the gas. Helium is the real gas closest to ideality, obviously because of its noble nature leading it to be not substantially attractive or repulsive, and because of its small single atom. Real gas attraction also has far more to do with size than molecular polarity, although obviously when gases are of comparable size more polar gases attract more than less.

Gas	Chemical formula	*a* (L^2 atm mol^{-2})	*b* (L mol^{-1})
Ammonia	NH_3	4.17	0.0371
Argon	Ar	1.34	0.0320
Carbon dioxide	CO_2	3.61	0.0429
Chlorine	Cl_2	6.25	0.0542
Fluorine	F_2	1.15	0.0290
Helium	He	0.0341	0.0238
Hydrogen	H_2	0.242	0.0265
Methane	CH_4	2.27	0.0430
Neon	Ne	0.205	0.0167
Nitrogen	N_2	1.39	0.0391
Oxygen	O_2	1.36	0.0319
Water	H_2O	5.46	0.0305

Table 3: van der Waals coefficients for selected gases. More complete data in the corresponding LibreTexts reference table.

Again, the corrections for a gas that has a low pressure and a high temperature are trivial. The major differences that you find between the ideal gas law and the van der Waals equation are computations that involve high pressure or those that involve low temperature.

Let's return to our example at the start. The ideal gas law gave us no way to distinguish 31.25 moles of oxygen from 31.25 moles of any other gas. The van der Waals equation, on the other hand, requires us to use values for *a* and *b* that are unique to oxygen, and allow us to correct for the size of the oxygen molecule and the interactions that happen between oxygen molecules. When we use the van der Waals equation to solve for the pressure inside this gas tank, we get an adjustment to the value we computed with the ideal gas law:

$$\left(P+\frac{n^2a}{V^2}\right)(V-nb)=nRT\rightarrow P=\frac{nRT}{V-nb}-\frac{n^2a}{V^2}$$

$$\rightarrow P=\frac{(31.25\text{mol})(0.082058\text{L atm mol}^{-1}\text{K}^{-1})(298\text{K})}{40.0\text{L}-(31.25\text{mol})(0.0319\text{L mol}^{-1})}-\frac{(31.25\text{mol})^2(1.36\text{L}^2\text{atm mol}^{-2})}{(40.0\text{L})^2}$$

$$\rightarrow P=19.6\text{atm}-0.830\text{atm}=\mathbf{18.8atm}$$

This isn't a massive correction! The uncorrected ideal gas computation has less than a 2% error when compared to this "corrected" van der Waals calculation. But it does give a picture of how the attraction between oxygen molecules relieves the pressure that would be exerted on the container, because of the subtracting out of the *a*-term in the rearranged van der Waals equation.

As an exercise for yourself, repeat this both the ideal gas computation and the van der Waals compuation for 1.000 gram of oxygen gas in this container. This should give you a very low pressure in the container (0.0191 atm) that isn't corrected for by the van der Waals equation at any significance; this will further emphasize that there's need to correct for the ideal gas law when pressures are high, not when pressures are low.

1.6: Another way of dealing with real gases: the virial equation

The van der Waals equation is an improvement on the ideal gas law, but it still has weaknesses. More sophisticated equations (Redlich-Kwong and Peng-Robinson are the two most prominent) use the same basic underpinning of correcting the ideal gas law with terms that represent molecular volume and molecular attraction.

On the other hand, a rawer mathematical approach to describing deviations from ideality is to express those deviations as part of a polynomial series where the constants have less intentional physical meaning, but instead are simply fine-tuned corrections. These types of approaches are called *virial equations*, so named as they are expressions of the force between molecules.

The most fundamental of the virial equations multiplies the right side of the ideal gas law by a power series:

$$PV = nRT\left(1 + \frac{nB}{V} + \frac{n^2C}{V^2} + \frac{n^3D}{V^3} + \cdots\right)$$

In this equation, B, C, and D (and all the values that would follow) are dubbed *virial coefficients*. In most practical applications, even a mere single correction using B, also known as the *second virial coefficient*, is sufficient:

$$PV = nRT\left(1 + \frac{nB}{V}\right)$$

Use of the third virial coefficient C and the fourth virial coefficient D, let alone further coefficients, are reserved only for specialized technical applications. (The first virial coefficient is in the first term of the parenthetical correction - simply 1 in all conditions.)

Gas	**If *T* = 100 K, B =**	**If *T* = 273 K, B =**	**If *T* = 373 K, B =**
Argon	-0.1870 L mol^{-1}	-0.0217 L mol^{-1}	-0.0042 L mol^{-1}
Helium	0.0114 L mol^{-1}	0.0120 L mol^{-1}	0.0113 L mol^{-1}
Hydrogen	-0.0020 L mol^{-1}	0.0137 L mol^{-1}	0.0156 L mol^{-1}
Neon	-0.0060 L mol^{-1}	0.0104 L mol^{-1}	0.0123 L mol^{-1}
Nitrogen	-0.1600 L mol^{-1}	-0.0105 L mol^{-1}	0.0062 L mol^{-1}
Oxygen	-0.1975 L mol^{-1}	-0.0220 L mol^{-1}	-0.0037 L mol^{-1}

Table 4: Second virial coefficients for selected gases as a function of temperature.
Data directly from the corresponding table in Monk's Physical Chemistry, with units converted.

Note that while B has units of L mol^{-1}, it is *not* the same as the van der Waals constant b and does *not* represent volume of molecules; this constant is an attempt to combine all the information included in the van der Waals a and b constants into a *single* value. For that reason, the meaning behind B is far more dependent on temperature than the van der Waals constants. The value of B for oxygen at 273 K is -0.0220 L mol^{-1}; however, that value approaches negligibility (at -0.0037 L mol^{-1}) at 373 K, and becomes far more significant (-0.198 L mol^{-1}) at 100 K.

It's not listed in the table above, but the value of B for oxygen at 298 K is -0.0158 L mol^{-1}. If we were to determine the volume correction for our kilogram (or 31.25 mol) of oxygen gas at standard ambient temperature, a solution of the simplified virial equation would give:

$$PV = nRT\left(1 + \frac{nB}{V}\right) \rightarrow P = \frac{nRT}{V}\left(1 + \frac{nB}{V}\right)$$

$$\rightarrow P = \frac{(31.25\text{mol})(0.082058\text{L atm mol}^{-1}\text{K}^{-1})(298\text{K})}{40.0\text{L}}\left(1 + \frac{(31.25\text{mol})(-0.0158\text{L mol}^{-1})}{40.0\text{L}}\right)$$

$$\rightarrow P = 19.1\text{atm}(1 - 0.0123) = \mathbf{18.9\ atm}$$

This is a correction very similar to that given by the van der Waals equation.

Because of the dependence on temperature that the virial coefficients show, virial coefficients are not traditionally tabulated simply. Papers and entire volumes are given over to the calculation of virial coefficients at a range of specific temperatures, and those coefficients are generally derived through gas behavior or empirically determined. Reliable approximations for many applications are genuinely valuable, and worth the price of a scientific text.

It's worth a mention, at the end of this particular discussion, that the quantity *n / V* turns up *repeatedly* in these equations, in the virial equation in particular. It's convenient to simplify this quantity, and we usually do so by creating a definition called the *molar volume* of a gas:

$$\overline{V} = \frac{V}{n} \text{units of} \overline{V} : \mathrm{L\,mol}^{-1}$$

This definition allows us to simplify the virial equation reasonably well:

$$PV = nRT\left(1 + \frac{nB}{V}\right) \rightarrow P\frac{V}{n} = RT\left(1 + \frac{n}{V}B\right) \rightarrow P\overline{V} = RT\left(1 + \frac{B}{\overline{V}}\right)$$

We can also apply this definition to the van der Waals equation, and achieve a similar simplification:

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT \rightarrow \left(P + \frac{n^2 a}{V^2}\right)\left(\frac{V - nb}{n}\right) = RT$$

$$\rightarrow \left(P + \frac{n^2}{V^2}a\right)\left(\frac{V}{n} - b\right) = RT \rightarrow \left(P + \frac{a}{\overline{V}^2}\right)\left(\overline{V} - b\right) = RT$$

We will use this definition repeatedly to make our lives easier as we start to dig into the actual discipline of thermodynamics. Be aware that, for clarity's sake, I make a very deliberate nomenclature choice and write the molar quantity with a bar over the symbol. This choice will take other quantities that other textbooks are less clear about and help us understand more deeply.

1.7: Connecting the van der Waals and the viral equations: the Boyle temperature

Using the molar volume definition, we can rewrite the ideal gas law as a ratio that will always equal one:

$$\frac{P\overline{V}}{RT} = 1 (\text{assuming ideality of the gas})$$

For real gases, however, this ratio very rarely is equal to one. This ratio is a useful metric for deviation of a gas from ideality, and it appears often enough that we can give this ratio its own definition, Z, the *compressibility factor*:

$$Z = \frac{P\overline{V}}{RT} (\text{Z is unitless})$$

If Z is less than 1, the gas is more compact than we'd predict the corresponding ideal gas to be; if Z is greater than 1, the gas is more spread out than we'd predict the corresponding ideal gas to be. Graphs of compressibilities of gases are some of the first comparison graphs you tend to get exposed to in physical chemistry, and it's worth it to look closely at an example of such a graph and interpret it.

Compressibility factors for nitrogen as a function of pressure and temperature

Compressibility factors as a function of temperature and pressure for nitrogen gas.
Image taken from "Retired Pchem Prof" on Wikimedia Commons; CC BY-SA 4.0.

The most vivid differences in compressibility on this graph for nitrogen are found at a temperature of 220 K. At this comparatively low temperature, nitrogen molecules are able to attract one another more than you'd expect for an ideal gas even when the pressure the gas is under gets relatively high; therefore the gas tends to be more densely packed than ideal, and the compressibility is less than 1. However, when pressure gets high enough (approximately above 230 bar), repulsion between the gas molecules because of their volume takes over, and the gas molecules are more spread out than you would expect an ideal gas to be at that pressure; the compressibility is greater than 1.

At higher temperatures, as you'd expect, there is much less attraction between the gas molecules because the speed at which the molecules are moving don't allow for a great deal of attraction. Once the temperature reaches a certain level, at no point do the gas molecules appear to attract one another into a more dense packing at all; any substantial pressure causes the repulsive effects to have greater significance.

The virial equation is even more elegantly written in terms of the compressibility factor:

$$P\overline{V} = RT\left(1 + \frac{B}{\overline{V}} + \frac{C}{\overline{V}^2} + \frac{D}{\overline{V}^3} + \cdots\right) \rightarrow Z = \frac{P\overline{V}}{RT} = 1 + \frac{B}{\overline{V}} + \frac{C}{\overline{V}^2} + \frac{D}{\overline{V}^3} + \cdots$$

Our tabulation of the second virial coefficient in the last section made note that the values of B depend on temperature. There were certain values of B for low temperatures that were negative, and for high temperatures that were positive. The immediate implication is that there is a temperature where B goes to zero, and (if we continue to ignore the higher terms of the virial expansion as we have been), the compressibility factor reverts back to one:

$$Z = \frac{P\overline{V}}{RT_B} = 1 T_B \rightarrow \text{Boyle temperature}$$

This *Boyle temperature* is unique for every gas. The virial equation doesn't give a whole lot of clarity to what the Boyle temperature actually represents, however, other than a temperature where virial properties magically go to zero.

Making the van der Waals equation fit into the form of the compressibility factor is a bit more of an algebraic challenge, but a worthwhile one to take on. Let's start by taking the van der Waals equation in terms of the molar volume, and solve the equation for pressure:

$$\left(P + \frac{a}{\overline{V}^2}\right)\left(\overline{V} - b\right) = RT \rightarrow P + \frac{a}{\overline{V}^2} = \frac{RT}{\overline{V} - b} \rightarrow P = \frac{RT}{\overline{V} - b} - \frac{a}{\overline{V}^2}$$

By then multiplying every term by the molar volume divided by RT, we can then build an expression that's equivalent to the compressibility factor:

$$P\frac{\overline{V}}{RT}=\frac{RT}{\overline{V}-b}\frac{\overline{V}}{RT}-\frac{a}{\overline{V}^2}\frac{\overline{V}}{RT}\rightarrow Z=\frac{P\overline{V}}{RT}=\frac{\overline{V}}{\overline{V}-b}-\frac{a}{\overline{V}RT}$$

We have one term in this difference with molar volume isolatable in the denominator; in order to make this equivalent to the virial equation, we need to have the other term set up like this as well. It turns out that the best first step to do this is to transform the first term into a Taylor series expansion. We start by dividing top and bottom of that fraction by the molar volume:

$$\frac{P\overline{V}}{RT}=\frac{\overline{V}/\overline{V}}{\overline{V}/\overline{V}-b/\overline{V}}-\frac{a}{\overline{V}RT}\rightarrow\frac{P\overline{V}}{RT}=\frac{1}{1-b/\overline{V}}-\frac{a}{\overline{V}RT}$$

We now have the first term in the form 1/1-*x*, which can be expanded by the Taylor series into $1 + x + x^2 + x^3 + \ldots$ Here, *x* is *b* divided by the molar volume:

$$\frac{P\overline{V}}{RT}=1+\frac{b}{\overline{V}}+\left(\frac{b}{\overline{V}}\right)^2+\left(\frac{b}{\overline{V}}\right)^3+\cdots-\frac{a}{\overline{V}RT}$$

Recall the full form of the virial equation that we rederived earlier in the section in terms of the molar volume:

$$\frac{P\overline{V}}{RT}=1+\frac{B}{\overline{V}}+\frac{C}{\overline{V}^2}+\frac{D}{\overline{V}^3}+\cdots$$

We've only provided values for the second virial coefficient *B*; that's because we made the decision that only extending the virial equation to the second term would be sufficient for approximating the real behavior of a gas. For a similar reason, we're going to limit the Taylor series expansion to eliminate terms in the molar volume squared and beyond. The simplified version of the rearranged van der Waals and virial equations, therefore, are these:

$$\frac{P\overline{V}}{RT}=1+\frac{b}{\overline{V}}-\frac{a}{\overline{V}RT}=1+\frac{b-a/RT}{\overline{V}}$$

$$\frac{P\overline{V}}{RT}=1+\frac{B}{\overline{V}}$$

The upshot of all of this rearrangement is that we've expressed the second virial coefficient *B* in terms of the two van der Waals coefficients:

$$B=b-\frac{a}{RT}$$

That's not the only benefit to this derivation. At the Boyle temperature T_B, we achieve ideality in gas behavior; in terms of the virial equation, $B = 0$ at the Boyle temperature. Therefore we can do one more simple rearrangement:

$$0=b-\frac{a}{RT_B}\rightarrow b=\frac{a}{RT_B}\rightarrow T_B=\frac{a}{bR}$$

The van der Waals constants, therefore, contain information about when the gas behaves ideally.

Building these kinds of theoretical connections between relationships is one of the core tasks we pursue in our study of physical chemistry. We can make a great deal of headway with algebraic rearrangement. Ultimately, however, we'll need the tools of the calculus to describe changes in the systems we'll study over time.

CHAPTER OVERVIEW

2: State functions, process functions, and the first law

Energy is not created or destroyed; it is merely transformed from one form to another.

When you take a course that's called *physical chemistry*, you've heard that statement a ton of times over the course of your life, in a ton of contexts. You may even recognize that a lot of those contexts are quite imprecise, and they don't tell you a whole lot.

Physical chemistry, however, is a discipline that requires precision. We don't want to use the language that describes energy imprecisely. How we classify the types of energy we will use to describe the physical systems we study is critical work, and we want to take that work seriously.

You're already familiar with two very specific kinds of energy; we know the phrases *kinetic energy* and *potential* energy very well. We instinctively associate kinetic energy with the motion of objects; it's worth it to think about what kinds of motion go into the specific behavior of gas molecules. That motion doesn't merely include moving back and forth in more-or-less straight line trajectories - the motion we call *translation* - but it involves the tumbling of molecules beyond the monoatomic, *rotation*, and also the wiggling and twisting of the bonds within those molecules - *vibration*. The fact that monoatomic gas particles can't undergo rotation and vibration the way that polyatomic gases can turns out to have massive implications in other energy changes. We'll deal with those later.

Gas molecules also have potential energy. Again, this could be the simple attraction or repulsion between particles. But this could also be the energy stored within bonds of a molecule, as that molecule starts to pull apart and finds itself pulled back together again. For both kinetic and potential energy, it's not useful to apply a single equation like $\frac{1}{2}mv^2$ or $\frac{1}{2}kx^2$ to the state of an atom or a molecule - we have to study the *whole range* of the behavior of the molecule to get its energy state.

And, you might imagine very quickly, that's complex to do for a single molecule. We *really* don't want to do this, molecule by molecule, for an entire container full of literally of billions of billions of molecules.

The power of how we construct the discipline of thermodynamics is found in how we build up quantities that take the properties of individual atoms and molecules and utilize them as descriptions of the much larger whole. Our first definition is of a quantity that takes all of the individual atoms and molecules and adds up all of the kinetic energies and potential energies - all of the translation, rotation, vibration, electric repulsion and attraction, and elastic stretching and compressing. That summed total energy for a specific system is what we will declare that system's *internal energy*.

What can we do to a system to change its internal energy - to add energy to the molecules in the system in all different ways? There are a host of suggestions that we can make, but we will find as we study the possibilities that they can be summarized in one of two ways - as state functions, or as process functions.

2.1: Heat and internal energy; state vs. process

Once again, it must be said and repeated: thermodynamics is a discipline that requires precision in our language. And so we want to give time to making sure we're precise with the word *system*.

The image at right (from the public domain on Wikimedia Commons) is a generic representation of a system. It is enclosed by a boundary; is has surroundings that are on the other side of the boundary. Here and often throughout our study of thermodynamics, we'll divide the universe into a system and its surroundings, and we'll study both independently of one another.

Systems can be open or closed. If a system is *open*, matter can get across the boundary; if a system is *closed*, all matter remains within the boundary. Even if the system is closed, though, energy can get across the boundary - energy in the form of work and/or heat. We'll discuss both kinds of energy here shortly.

It is a very special sort of system that doesn't allow even energy to get across the boundary; we call that kind of closed system an *isolated* system.

For the most part, as we study systems, we're not going to be interested in open systems; the distinction between the system and the surroundings gets fuzzy in that case. We're not going to be interested much in isolated systems either; problems where no energy is transferred are boring problems. It's the closed system that allows energy back and forth across the boundary where most of the interesting concerns of thermodynamics are found. Systems of that sort are what we are referring to below.

One way to add energy to a system is to raise its temperature, and the simplest way can envision doing so is bringing that system into contact with something at the higher temperature, and letting the two systems sit together for a while, with energy crossing the boundary between the two systems, until the temperatures equal out. The fact that this happens is something that is so instinctive and natural to our experience that we can call it a scientific law. It's a somewhat silly law, so silly that we give it the silly name *the zeroth law of thermodynamics*, but it's worth mentioning because without it, we don't have a useful definition of temperature.

> ***The zeroth law of thermodynamics: if system A and system B have the same temperature, and system B and system C are at the same temperature, then system A and system C must also have the same temperature.***

Knowing we can define a given temperature this way does two things. First, it makes sure we have a definition of thermometers that works; it's a device that you can bring into *thermal equilibrium* with one stuff and know that it's also in thermal equilibrium with a second stuff if it gives you the same reading.

The second thing it does is assemble the terms we'll use to describe the transfer of *heat*. Energy has to move from one system to another if those systems are going to achieve the same temperature. If all other state quantities remain constant, when two systems of different temperature come into contact with one another, heat energy is released from the system at higher temperature (or the *hot* system) and is absorbed by the system at lower temperature (the *cold* system).

The classic definition of heat is a quantity of energy that is directly proportional to temperature change. The constant of proportionality is the *heat capacity* of a system, which is a measure of how much that system resists temperature change:

$$q = C\Delta T \text{units of} C \text{: J K}^{-1}$$

It turns out that if volume is held constant, all of this heat transferred will *always* be the only source of internal energy change for the system, and we can make this expression exactly equivalent:

$$\Delta U = C_V \Delta T \text{units of} C_V \text{: J K}^{-1}$$

The "*V*" subscript on the heat capacity symbol C_V represents volume of the system being held constant - this is the most common kind of heat capacity, but not the only kind, especially for gases! We'll expand our picture of heat capacities later.

What's more interesting in this moment, and will have more implications for how we study energy in the long run, is the difference in how we represent heat (q) and how we represent internal energy change (ΔU), which under conditions of constant volume are equivalent properties. Notice that the symbol we use for both are different, because the way the energies are accounted for are different.

Internal energy change is the sum of kinetic and potential energy changes with the system. Each of these quantities are *state energies* - they are energies that describe the specific state of the system at a given point in time. A system can possess internal energy, because individual molecules can possess a given kinetic and potential energy at a moment in time.

Heat is different; we *never* describe a system as possessing heat. Heat is a *process energy*, or a property of the transfer of energy between two systems, or between a system and its surroundings. When a hot system transfers heat to a cold system, we say both that the internal energy of each system has changed - the internal energy of the hot system decreases, and the internal energy of the cold system increases. We also say that heat has been transferred between the two systems - the hot system releases heat, and the cold system *absorbs the same heat.*

Process energies are often called *path energies* in the language of thermodynamics, and that's because the energies are considered to be functions of the path required to change a condition from initial to final. If you consider the two paths at right up Mount Kilimanjaro (image taken from OpenStax Chemistry, CC BY 4.0) it's quite plain that the direct path X will require less work (although possibly more agility!), while path Y is longer and will require more work. But when you take the difference in potential energy between the two paths, you get the same result; you started from the same low elevation regardless of path, and you ended at the same high elevation. Potential energy is a *state energy*; the change in potential energy only depends on the initial and final points. Work is a *process energy*; it depends on the path taken. (In a thermodynamics context, we'll say more about work later on.)

So, again, internal energy is a state energy; heat is a process energy. This difference in how we define internal energy against how we define heat also informs the distinction in the symbol set. Like kinetic and potential energy, internal energy can have initial and final values; the difference between initial and final internal energies is represented by the classic Greek capital-delta, Δ. *There is no such thing as change in heat; there is no such thing as* Δq, *and you should never write such a quantity.* When a system is held at constant pressure, the change in internal energy is *equal to* the heat transferred; we write $\Delta U = q$.

The heat capacity C_V is a classic *extensive* property - a property of an object or a system alone. If we want to make sure we're studying a material *intensively*, what we're taught to do in general chemistry coursework is to divide the heat capacity by the mass of the homogenous system, and determine that object's *specific heat*:

$$q = mc\Delta T \text{units of} c\text{: J kg}^{-1}\text{K}^{-1}$$

In physical chemistry, however, we find it more useful to focus on not a specific heat dictated by a material's mass, but a *molar specific heat*:

$$q = n\overline{C}\Delta T \text{units of} \overline{C}\text{: J mol}^{-1}\text{K}^{-1}$$

$$\Delta U = n\overline{C_V}\Delta T \text{units of} \overline{C_V}\text{: J mol}^{-1}\text{K}^{-1}$$

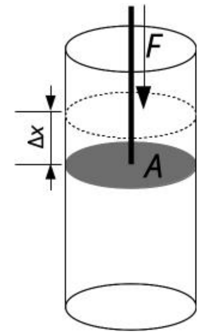
The bar over the quantity $\overline{C}$ (and its constant-volume counterpart $\overline{C_V}$), as we previously did with *molar* volume, is our tell that we are using a *molar* variety of the quantity. This gets designated with subscripts in other textbooks (say, as $C_{V,m}$) and sometimes the designation of molar quantities is inconsistently done, as we will point out in coming sections. We will make every effort in this text to be consistent with our use of bars to designate molar quantities.

2.2: Work and the integral; constant vs. variable pressure

We have a definition of work already set:

$$w = F\Delta x\cos\theta (\text{single constant force acting on single object})$$

This definition sets a sign convention for work, as we've said before; forces in the same direction as an object's displacement result in positive work done, and forces in the opposite direction as the displacement yield negative work; we can characterize positive work as work done on a system and negative work as work done by a system in this way. Like heat, work is a process energy; no system possesses work, but work is performed on a system.



Consider the system to be a gas in a closed container, as shown above and at right; the force is applied to the gas by a piston of area A, which is displaced a distance of Δx. Knowing the area of the piston allows us to do two things: we can recognize pressure in the force per unit area, and we can also recognize the change of volume in the system from the product of the area and the displacement. This has the effect of multiplying the work relation by 1:

$$w = \frac{F}{A}(A\Delta x) = -P_{ext}\Delta V (\text{single constant force applying pressure on a gas system})$$

The pressure here is that which is applied from the outside of the gas; the positive work results from the motion of the piston in a way that decreases the volume of the gas, which introduces a negative sign to the work relation. To do work on a gas, you must decrease its volume; to have work done by a gas, the volume of the gas must increase.

The mathematics here is very straightforward as long as the pressure applied externally to the gas is constant. Realistically, that's not going to be the case; usually, some kind of increase in pressure is going to be necessary to start the piston moving, and if the piston moves rapidly that will require a pressure that constantly changes. Even studying the ideal gas law solved for pressure ($P = nRT / V$) provides a model of the number of different ways that pressure can change as a function of the other terms of the variables of state.

Calculus gives us tools to study the transition from a determination that's dependent on a constant quantity to a determination that's dependent on a variable quantity. The first step of building this extension is to change the quantities w and ΔV from discrete quantities to quantities that have differential size - very very small in nature:

$$\delta w = -PdV$$

Then what happens is *integration* - adding up the very small changes over the range of the transition:

$$\int_{process} \delta w = -\int_{V_i}^{V_f} PdV$$

The quantity δw is what is known as an *imperfect differential*. You can't integrate something from initial-work to final-work when work isn't a property of state, merely a property of a process. You *can* add up the individual differential processes that lead to the whole. So an integral that looked like "w δw" will never be performed; integrating δw will add up to the total work done:

$$w = -\int_{V_i}^{V_f} PdV$$

dV, however, is a proper differential and can be integrated over traditionally. If P is a constant (that is, if the process of changing the volume of the gas is *isobaric*), dictated by the external pressure, P can be simply removed from the integral in order to regenerate the relationship we had before:

$$w = -P_{ext}\int_{V_i}^{V_f} dV = -P_{ext}\Delta V$$

This isn't all that interesting.

But we can also use this opportunity to consider what it looks like when moles and temperature are held constant - that is, the container is closed and the process of changing the volume is *isothermal*. In that case, we can replace P with a term that's solved out of the ideal gas law:

$$P = \frac{nRT}{V} \rightarrow w = -\int_{V_i}^{V_f} \frac{nRT}{V} dV$$

(There's an implied assumption in this substitution - we are asserting that the external pressure applied as the volume changes is equivalent to the pressure inside the container. This should make sense on the surface - if we think about this change of volume as a compression, then the pressure required to continue the compression is going to increase as the volume of the gas gets smaller. But this actually demands a more subtle implication to be accurate as a integration - that the change in volume is slow, gradual, and *reversible*. Practical attempts to make isothermal changes in volume are not reversible. We won't bother ourselves too much with this distinction right now - but we will have to think about it later.)

This integration isn't as direct as it would be if the pressure were constant; it requires the taking of an antiderivative, in this case the antiderivative of $1/V$:

$$w = -nRT \int_{V_i}^{V_f} \frac{1}{V} dV = -nRT[\ln V]_{V_i}^{V_f} = -nRT \ln \frac{V_f}{V_i}$$

At any point in this process, we could have chosen to get rid of the negative sign by swapping the limits on the integral, so this is an equivalent expression:

$$w = nRT \ln \frac{V_i}{V_f}$$

All of these expressions dictate a sign convention for work that turns what we expect in signs on its head. If the volume of the gas we're studying in this particular example is *decreasing*, the sign of work that results is *positive*. If the volume is increasing, the volume that results is negative.

This might seem backwards. But it's more satisfying to consider the signs in terms of how they impact the energy of the system as a whole.

> ***The first law of thermodynamics: the two ways to change the internal energy of a system are first through the release or absorption of heat, and second through the performing of work, either on the system or by the system.***

$$\Delta U = q + w$$

	If the sign is *positive*...	**If the sign is *negative*...**
q	then the system's temperature *increases* then the system *absorbs* heat	then the system's temperature *decreases* then the system *releases* heat
w	then the system's volume *decreases* then work is done *on* the system	then the system's volume *increases* then work is done *by* the system
ΔU	then the total energy of the system *increases*	then the total energy of the system *decreases*

Both positive values of heat and work correspond to energy being added to the system in question. If heat transfer is positive, then heat has been absorbed by the system, causing the total energy of the system to increase. If work is positive, then work has been done on the system, causing the total energy of the system to increase.

We're making important assertions with these terms of our sign conventions. Strictly speaking, it is possible for a system to absorb or release heat without changing temperature; consider a phase change, for example. However, for gas systems, we can assert with confidence that if the temperature does not change, there's been no heat absorbed or released, and any amount of internal energy change comes from work done. And we can also assert that if the volume of a system does not change, no work is done on or by the system, and any internal energy change comes from heat.

This will be critically important as we build the key definitions that surround our measurement of heat.

2.3: Specific heats of ideal gases

We have already established a definition of temperature from earlier in the course:

$$T = \frac{2K_{tot}}{3Nk_B}$$

If we rearrange this definition for the total kinetic energy of molecules, we have a first view at what an expression for internal energy is going to look like:

$$K_{tot} = \frac{3}{2}Nk_BT$$

This would approximate the internal energy of a *monoatomic* ideal gas, because the amount of ways you can store energy in a gas like helium or neon is so limited; those “molecules” (which, of course, are mere single atoms) can’t rotate or vibrate like even hydrogen gas can.

k_B, you’ll remember, is the constant equivalent to 1.381 × 10^{-23} J K^{-1}, the Boltzmann constant (to use the common value and not a value with significant figure overkill). This turns out to be a reference energy *per molecule*; if you multiply this constant by Avogadro’s number, you get the ideal gas constant *R*, which is 8.3145 J mol^{-1} K^{-1}.

If we transform the number of molecules times a molecular reference energy to a number of *moles* times a *molar* reference energy - which is what the ideal gas constant turns out to be! - and we presume that this kinetic energy is equal to internal energy for an ideal gas of single atoms, this is what results:

$$U = \frac{3}{2}nRT$$

Other texts use this as the foundational definition of internal energy for a monoatomic system. They even ascribe a meaning to the “3” in the numerator - the three dimensions, *x*, *y*, and *z*, that an ideal gas atom can move about in.

We’ve already argued that if there’s no volume change in a gas, there’s no work done on or by that gas. One of the key outcomes of that statement is that, under circumstances of constant volume, the change in internal energy is exactly equivalent to the heat absorbed.

That statement makes *both* of these statements true:

$$q_V = \Delta U = \frac{3}{2}nR\Delta T (\text{presuming monoatomic ideal gas})$$

$$q_V = \Delta U = n\overline{C_V}\Delta T (\text{true in general})$$

The equivalence of these two statements can be simplified into a single term, which we’re going to use to define a brand-new specific heat:

$$\overline{C_V} = \frac{3}{2}R = 12.47 \text{J mol}^{-1}\text{K}^{-1} (\text{presuming monoatomic ideal gas})$$

The monoatomic ideal gas constant-volume specific heat $\overline{C_V}$ is one of the more remarkable theoretical results - the first four periodic gases in the periodic table all have molar specific heats of 12.5 J mol^{-1} K^{-1} under conditions of constant volume, and deviations for the larger ideal gases are minor and only in the third significant figure (xenon’s value for the molar constant-volume specific heat is 12.7 J mol^{-1} K^{-1}; it’s larger owing to size and intermolecular attraction).

Even more remarkable are the patterns this reveals about gas behavior.

Let’s think about reality for a bit.

We need to study gases in containers of constant volume. When we receive gases to study in a lab, they’re typically shipped in steel tanks, and the volume of those tanks isn’t going to shift. Rigid containers are a pretty important part of the story of our study of gases.

They’re also not the conditions we normally experience on a day-to-day basis. The rooms we study in are open, and matter can move in those rooms freely. To the extent that we might generate gas in the lab, that gas typically is collected over water - hardly a constant-volume situation, given the capacity we have to displace water.

Behavior we want to model that will have real, practical utility for us isn't behavior at constant volume, but at constant *pressure*. Pressure is far less likely to change than volume. The atmosphere keeps the pressure we experience in a lab setting, if not perfectly constant, then reasonably close to constant.

The bad news is that we have to deal with work in that circumstance. The good news is that the work is taking place (by definition!) at constant pressure, so that we don't have to deal with an integral variety of work; we can simply make the statement that $w = -P\,\Delta V$.

What this will result in is a new, *different* specific heat. Now, internal energy change and heat transfer are *not* equal; the presence of work changes one with respect to the other. What is most interesting to do in this case is to actually solve for a heat transfer at constant pressure - call this q_P, as a counter to q_V - by rearranging the first law:

$$q_P = \Delta U - w$$

Note that subtracting work flips the sign on $P\,\Delta V$:

$$q_P = \frac{3}{2} nR\Delta T + P\Delta V (\text{presuming monoatomic ideal gas})$$

It should be somewhat obvious here that applying the ideal gas law to $P\,\Delta V$ can make this relationship easily simplified if we work on the assumption that this is a closed container where the count of particles in the container cannot change, and that changing the volume of this matter (likely a gas) at constant pressure will only serve to change the temperature:

$$P\Delta V = nR\Delta T (\text{assuming constant pressure and moles})$$

$$q_P = \frac{3}{2} nR\Delta T + nR\Delta T (\text{presuming monoatomic ideal gas})$$

$$q_P = \frac{5}{2} nR\Delta T$$

Once again, we can recognize that there is a molar specific heat embedded within this expression, and we can separate that specific heat out to identify that this is different than what holds for matter at constant volume:

$$\overline{C_P} = \frac{5}{2} R = 20.79 \text{J mol}^{-1} \text{K}^{-1} (\text{presuming monoatomic ideal gas})$$

Again, for most noble gases, there's spectacular agreement with this ideal value of constant-pressure specific heat $\overline{C_P}$ to three significant figures.

We therefore have the key difference in the molar specific heats of gases. Gases at constant volume have a lower specific heat than gases at constant pressure. What's more…

$$\overline{C_P} - \overline{C_V} = \frac{5}{2} R - \frac{3}{2} R (\text{presuming monoatomic ideal gas})$$

$$\overline{C_P} - \overline{C_V} = R (\text{true in general})$$

...that difference is *exactly* the ideal gas constant. And while we've done all of this derivation for monoatomic ideal gases, because that difference emerges from the addition of the work term in pursuing $\overline{C_P}$, *the difference between molar specific heats at constant pressure and at constant volume will always be equal to the ideal gas constant.*

Diatomic ideal gases, with rotational and vibrational degrees of freedom to store internal energy (in addition to translational degrees of freedom), have higher values of the constant-pressure and constant-volume specific heats:

$$\overline{C_P} = \frac{7}{2} R (\text{presuming diatomic ideal gas})$$

$$\overline{C_V} = \frac{5}{2} R (\text{presuming diatomic ideal gas})$$

The difference between those two values for diatomic gases - and for any ideal gases, for that matter - will remain *R*.

(A quick footnote: The textbook I was using to teach physical chemistry when I started working on these notes, which will remain nameless to protect the guilty, *consistently* switched the V and P subscripts when identifying the constant-pressure and constant-volume specific heats, in 14 different instances throughout the second chapter of the text, despite the ease with which it can be demonstrated that the constant-pressure specific heat should *always* be greater than the constant-volume specific heat for a gas. As you get more advanced in your chemical education, these types of typographical errors *will* turn up in textbooks, and you need to be aware of the possibility of them and work through the theory for yourself to demonstrate that the text you're studying from is accurate.)

2.4: Adiabatic processes - energy change without heat transfer

In establishing the first law of thermodynamics, we've stated that all of the changes in the internal energy of a system can come from one of two places - heat transfer across the system boundary, and work performed on or by the system. We've stated further that this internal energy change in the system is a *state energy* - that it is only dependent upon the initial and final conditions of the system.

Those statements mean that this statement (which we proposed in the previous section) is true in general:

$$\Delta U = n\overline{C_V}\Delta T$$

This represents, of course, the heat transfer at constant volume. But because the work done by the system is zero when the volume is held constant, it *also* represents the internal energy change - and, because the internal energy is a state energy, that means that this statement represents the internal energy change under *any* condition, whether the volume is held constant or not.

It can even represent the internal energy change *when no heat is transferred across the system boundary at all*.

Processes where no heat is transferred are called *adiabatic* processes, and these special cases have a host of interesting consequences.

In an adiabatic process, the entirety of the internal energy change in a system is a result of work performed. It's worthwhile to frame both the internal energy changes and the work performed in terms of differentials, rather than full deltas, in order to consider the full range of changes to the system. Let's first rewrite the internal energy change as a differential change:

$$dU = n\overline{C_V}dT$$

We can remind ourselves of the differential definition of work:

$$\delta w = -PdV$$

The differential statement of the first law of thermodynamics is an equation of the inexact differentials of heat and work (because heat and work are process energies) to the exact differential of internal energy:

$$dU = \delta q + \delta w$$

Because δq is equal to zero in an adiabatic process (if no heat energy is transferred across the boundary, not even differential heat counts), dU equates to δw, and we have a new equation:

$$dU = \delta w (\text{adiabatic process})$$

$$n\overline{C_V}dT = -PdV (\text{adiabatic process})$$

Now this is a relationship that is describing the differential change in two state variables - temperature and volume. By placing this in terms of differentials, we will be able to integrate both sides of the relationship with appropriate rearrangement, and arrive at a relationship that will describe how volume changes with changing temperature in an adiabatic process. Let's assume our system is an ideal gas for simplicity, and let's make the rearrangements to make this equation integrable.

Because we're assuming an ideal gas, we can replace P on the right side:

$$n\overline{C_V}dT = -\frac{nRT}{V}dV (\text{adiabatic process})$$

The moles of the gas cancel, and we can easily divide both sides by T to gather like terms on the same side of the equation:

$$\frac{\overline{C_V}}{T}dT = -\frac{R}{V}dV (\text{adiabatic process})$$

We've acknowledged before that the antiderivative of $1/V$ is $\ln V$, and T doesn't behave any differently. We can then perform an integration and easily rid ourselves of an annoying negative sign:

$$\int_{T_i}^{T_f} \frac{\overline{C_V}}{T}dT = \int_{V_i}^{V_f} -\frac{R}{V}dV (\text{adiabatic process})$$

$$\overline{C_V}[\ln T]_{T_i}^{T_f} = -R[\ln V]_{V_i}^{V_f} \text{ (adiabatic process)}$$

$$\overline{C_V}\ln\left(\frac{T_f}{T_i}\right) = -R\ln\left(\frac{V_f}{V_i}\right) \text{ (adiabatic process)}$$

$$\overline{C_V}\ln\left(\frac{T_f}{T_i}\right) = R\ln\left(\frac{V_i}{V_f}\right) \text{ (adiabatic process)}$$

This relationship is powerful enough, but we can simplify it further. Remember that *R* has the same units as a molar specific heat, and further that the difference between the specific heats $\overline{C_P}$ and $\overline{C_V}$ is exactly equal to *R*. Therefore we can set up a very interesting ratio on the right-hand side:

$$\ln\left(\frac{T_f}{T_i}\right) = \frac{R}{\overline{C_V}}\ln\left(\frac{V_i}{V_f}\right) \text{ (adiabatic process)}$$

$$\ln\left(\frac{T_f}{T_i}\right) = \frac{\overline{C_P} - \overline{C_V}}{\overline{C_V}}\ln\left(\frac{V_i}{V_f}\right) \text{ (adiabatic process)}$$

$$\ln\left(\frac{T_f}{T_i}\right) = \left(\frac{\overline{C_P}}{\overline{C_V}} - 1\right)\ln\left(\frac{V_i}{V_f}\right) \text{ (adiabatic process)}$$

We make this final rearrangement here because it's common, in the study of ideal gas relationships, to define the ratio between constant-pressure and constant-volume specific heats with a symbol of its own, *γ*:

$$\gamma = \frac{\overline{C_P}}{\overline{C_V}}$$

That ratio makes the statement of this relationship between changing temperature and changing volume easier to express:

$$\ln\left(\frac{T_f}{T_i}\right) = (\gamma - 1)\ln\left(\frac{V_i}{V_f}\right) \text{ (adiabatic process)}$$

There's one more simplification we can make to this expression. It's a basic property of a logarithm that if a constant *c* is multiplied by a logarithm ln *x*, that the argument *x* of that logarithm can be raised to the *c* power:

$$c\ln x = \ln x^c$$

We can apply this to the new constant (*γ* - 1) in our simplified expression:

$$\ln\left(\frac{T_f}{T_i}\right) = \ln\left(\frac{V_i}{V_f}\right)^{\gamma-1} \text{ (adiabatic process)}$$

At this point, the arguments of the logarithms, are equivalent, and the logarithms themselves are surplus to requirements:

$$\frac{T_f}{T_i} = \left(\frac{V_i}{V_f}\right)^{\gamma-1} \text{ (adiabatic process)}$$

This result is somewhat surprising if you've committed the gas laws that emerge from the ideal gas law to memory. We're very used to seeing this relationship, the rule known as Charles' Law:

$$\frac{T_f}{T_i} = \frac{V_f}{V_i} \text{(closed container at constant pressure)}$$

But, as the text to the right indicates, that statement isn't true without any context; we keep moles and pressure constant in the ideal gas law to arrive at such a relationship. We started this derivation indicating that pressure *was not* constant; it changed as a function of volume of the container. **Under adiabatic processes,** as a result, **when the volume of the gas decreases, the temperature of the gas increases.** This is a result of the temperature change being driven by work done on the gas, increasing the internal energy of the gas.

Derivations of this sort are important, and it is very easy to get lost in the mathematics and lose sight of the importance of the result. Both are important; the mathematics are the tools of our theoretical work, and it's important to use the mathematics well. But we're not blindly rearranging (much as it might feel like it sometimes!); we're rearranging to build simplified relationships that describe behavior under unique conditions.

2.5: A new state energy - enthalpy

The knowledge that more heat has to transfer into a gas to change its temperature under constant *pressure* than at constant *volume* also begs the introduction of a new quantity. We have already argued that the change in internal energy of a system is exactly the heat transferred between that system and its surroundings when that system is held at constant volume:

$$\Delta U = q_V$$

Let's define a new state energy - we will call it the *enthalpy* of a system, and we will give that energy the symbol H - to deal with the realities of the behavior of a system at constant pressure, rather than constant volume. We will define this quantity in this way:

$$H = U + PV$$

If we talk about how this state function changes over time, and we specify that this change will be under conditions of constant *pressure*, then we can write that change as

$$\Delta H = \Delta U + P\Delta V (\text{presuming constant pressure})$$

But $P\,\Delta V$ is nothing more or less than the negative of the work done on the system, which simplifies the expression to the point where we can apply the first law of thermodynamics:

$$\Delta H = \Delta U - w (\text{presuming constant pressure})$$

$$\Delta H = q_P$$

Therefore, while the change of internal energy is the heat transferred to that system at constant volume, we have defined this new quantity such that *the change of enthalpy of a system is the heat transferred to that system at constant pressure.* This is the connection between heat and enthalpy that has been talked about loosely ever since you took general chemistry. Now that we have defined and declared this definition formally, it is *necessary* that we keep this definition of enthalpy change specific; we will use it repeatedly going forward.

The introduction of the concept of enthalpy is where we move into the specific discipline of *chemical* thermodynamics from the general study of thermodynamics. Enthalpy is a concept that doesn't get addressed, or is only addressed in passing, in most general treatments of the transfer of heat, but the enthalpy concept is powerful in chemistry because it takes a transfer of heat under specific conditions - the most common laboratory conditions, constant pressure - and identifies it as a state energy. This makes the enthalpy concept supremely useful to the chemist, because it allows for the prediction of heat transfer from complex chemical reactions from other, more well-characterized chemical reactions.

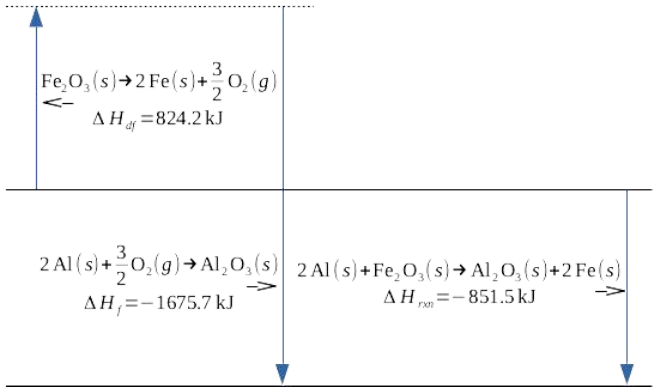
This is the conceptual description of the principle of chemistry known as *Hess' Law - if a series of chemical processes starts with materials A and ends with materials B, the enthalpy change for that series of chemical processes is equivalent to the enthalpy change corresponding with the direct reaction of materials A to materials B.* Let's give a specific example of this, the example of the famous thermite reaction:

$$2\mathrm{Al}(s) + \mathrm{Fe_2O_3}(s) \rightarrow \mathrm{Al_2O_3}(s) + 2\mathrm{Fe}(s)$$

This reaction is the sum of two other reactions, the *endothermic* deformation of iron(III) oxide to its constituent elements in their standard states, and the highly *exothermic* formation of aluminum oxide from its constituent elements in their standard states:

$$\mathrm{Fe_2O_3}(s) \rightarrow 2\mathrm{Fe}(s) + \frac{3}{2}\mathrm{O_2}(g) \Delta H_{df} = 824.2\mathrm{kJ}$$

$$2\mathrm{Al}(s) + \frac{3}{2}\mathrm{O_2}(g) \rightarrow \mathrm{Al_2O_3}(s) \Delta H_f = -1675.7\mathrm{kJ}$$



Formation reactions are well-characterized for a wide variety of products; any standard table of thermodynamic data is a good starting point for drawing this data. I'm using *deformation*, or *df*, to represent the decomposition of a reactant to its constituent elements in standard states; it's the opposite of a formation reaction; the enthalpy of deformation is the negative of the enthalpy of formation.

Determining an enthalpy of reaction for a process like the thermite reaction, then, involves taking the arithmetic sum of the enthalpies of deformation of the reactants and the enthalpies of formation of the products. This statement of Hess' Law is expressed in many general chemistry textbooks as simply taking the the sum of the enthalpies of formation of products, and subtracting from that the sum of the enthalpies of formation of the reactants:

$$\Delta H^\circ_{rxn} = \sum n\Delta H^\circ_{f,products} - \sum n\Delta H^\circ_{f,reactants}$$

The degree-symbol applied to the enthalpy symbol Δ*H*° represents the change in enthalpy at standard ambient temperature and pressure (SATP; 1 atmosphere and 298 K, or 25°C). This is implied frequently in general chemistry work, but we'll need to be explicit about the distinction between SATP and non-standard conditions later.

Take another reaction as an example, the combustion of propane gas in the presence of oxygen to products of carbon dioxide and water vapor:

$$\mathrm{C_3H_8}(g) + 5\mathrm{O_2}(g) \rightarrow 3\mathrm{CO_2}(g) + 4\mathrm{H_2O}(g)$$

This reaction has multiple moles of reactants and products, in addition to a quantity of oxygen in its standard state (and the enthalpy of formation of any element in its standard state is zero). The enthalpies of formation of the compound product as well as the reactants can be stated as molar quantites:

$$3\mathrm{C}(s) + 4\mathrm{H_2}(g) \rightarrow \mathrm{C_3H_8}(g)\Delta H^\circ_f = -103.8\mathrm{kJ\ mol^{-1}}$$

$$\mathrm{C}(s) + \mathrm{O_2}(g) \rightarrow \mathrm{CO_2}(g)\Delta H^\circ_f = -393.5\mathrm{kJ\ mol^{-1}}$$

$$\mathrm{H_2}(g) + \frac{1}{2}\mathrm{O_2}(g) \rightarrow \mathrm{H_2O}(g)\Delta H^\circ_f = -241.8\mathrm{kJ\ mol^{-1}}$$

The application of the equation form of Hess' Law to this reaction is therefore the sum of the enthalpies of formation of carbon dioxide and water vapor, each multiplied by their stoichiometric coefficients, subtracting the enthalpy of formation of propane:

$$\Delta H^\circ_{rxn} = 3(-393.5\mathrm{kJ\ mol^{-1}}) + 4(-241.8\mathrm{kJ\ mol^{-1}}) - (1)(-103.8\mathrm{kJ\ mol^{-1}}) = -2043.9\mathrm{kJ\ mol^{-1}}$$

The value of -2043.9 kJ $\mathrm{mol^{-1}}$ has to be interpreted carefully. The reactant with the stoichiometric coefficient of 1 is the propane itself, C_3H_8. Therefore the standard enthalpy of reaction for the combustion of propane is -2043.9 kJ of energy released per mole of propane combusted. This brings us to an inconsistency in our notation system: frequently enthalpies of formation and enthalpies of reaction are provided as molar quantities, and in the name of consistency ΔH°_{rxn} should be presented with a bar over it. This is rare to see in textbooks, however, and it's frequently simply assumed that ΔH°_{rxn} is a molar quantity without any further notation provided. We'll make more effort to be consistent in the future, but Hess' Law is one of the places where it's most commonly assumed that enthalpies are molar quantites.

Hess' Law is one of the primary applications of the enthalpy concept, and the application of the enthalpy concept most clearly connected to chemistry. But there are other applications of the enthalpy concept. At constant pressure, any heat transfer represented by q can be written as an enthalpy change ΔH. There are also values of enthalpy associated with *phase changes*, the transformation of matter from one phase to another. The *enthalpy of fusion* for water's transformation between liquid and solid states is ΔH_{fus} = 6.01 kJ mol^{-1} = 333.6 J g^{-1}; one mole of ice absorbs 6010 J of energy as it melts to water, and one mole of water releases 6010 J of energy as it freezes to ice. Likewise, the *enthalpy of vaporization* for water's transformation between gaseous and liquid states is ΔH_{vap} = 40.67 kJ mol^{-1} = 2.258 kJ g^{-1}.

2.6: Heat capacity and the partial derivative

Let's review the ideas within this chapter, and restate some principles a new way as we go.

All of the internal energy change in a system comes from one of two sources: heat transfer into or out of the system, and work done by or on the system.

$$\Delta U = q + w$$

If there's no volume change in the system, $w = 0$, and the internal energy change equates to heat transfer. This is the reason why we can equate the internal energy change to constant-volume heat capacity times temperature difference:

$$\Delta U = C_V \Delta T$$

This is true not merely for discrete changes that we represent by deltas, but for infinitesimal changes as well, so we can write this expression differently:

$$dU = C_V dT$$

This implies a derivative - but we have to be careful about how we write this derivative. The internal energy U is capable of changing when volume changes as well as when the temperature changes. When we write most simple derivatives in calculus, it's either directly stated or strongly implied that only one variable is changing as a function of the other. In thermodynamics, we're rarely constrained to a single variable changing, and we have to ensure that other variables are held constant.

Therefore it's necessary to express the derivative that's equivalent to C_V in a new way:

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

As we've referred to before, the quantity in parentheses is referred to as the *partial derivative* of internal energy with respect to temperature. The V outside of the parentheses means that, when taking this derivative of internal energy, *volume is held constant and not allowed to change the internal energy*; the derivative is based on the changes in temperature *alone.* In every other sense, taking this derivative is no different than any other derivative of a single variable we might take; but because internal energy is not merely based on possible changes in temperature, we have to *constrain* the derivative.

As you've seen in previous sections, though, we don't usually study the constant-volume heat capacity; we're more likely to study the constant-volume specific heat. So it's better to express this partial derivative in terms of molar properties:

$$\overline{C_V} = \left(\frac{\partial \overline{U}}{\partial T}\right)_V$$

Internal energy is equivalent to heat transfer at constant volume; we built the concept of enthalpy to be equivalent to heat transfer at constant pressure. So we can write an equivalent expression to equate to the constant-pressure specific heat:

$$\overline{C_P} = \left(\frac{\partial \overline{H}}{\partial T}\right)_P$$

(The difference between the quantity H and the quantity $\overline{H}$ is the difference in an energy presented in Joules or kiloJoules and an energy presented in Joules per mole or kiloJoules per mole. This has been implied previously; I'm making sure I say it explicitly here.)

There are several implications of this equivalence of the specific heats to partial derivatives of internal energy and enthalpy, respectively. One of these implications is that the specific heats are state variables, just like internal energy and enthalpy are. It turns out that we can apply Hess' Law to specific heats in the exact same way as we apply it to enthalpy itself:

$$\Delta\overline{C_P} = \sum n\overline{C_P}_{,products} - \sum n\overline{C_P}_{,reactants}$$

The idea that the specific heat of substances change when substances themselves chemically change should not be totally surprising, but it may build confidence to know that there's a clear pattern to *how* that specific heat changes.

There's an additional power to knowing how specific heat changes in a chemical reaction, and that's found in the connection between the enthalpy and the constant-pressure specific heat. We can undo the partial derivative just as easily as we can do it:

$$d\overline{H} = \overline{C_P}dT$$

We can integrate both sides of this equation and arrive at a change in enthalpy as a function of temperature discretely:

$$\int_{H_1}^{H_2} d\overline{H} = \int_{T_1}^{T_2} \overline{C_P}dT$$

$$\overline{H}_{T_2} - \overline{H}_{T_1} = \int_{T_1}^{T_2} \overline{C_P}dT$$

Now, we could just rewrite the left side as $\Delta\overline{H}$. However, that's not a typical enthalpy change *chemically*; we prefer to write enthalpy changes not for simply transitioning the temperature of a single known stuff, but for the chemical transformation of one group of substances to a new group of substances. This corresponds to $\Delta\overline{C_P}$ that we just defined above. It's customary to rewrite this new equation in terms of the changes in reaction of these substances:

$$\Delta\overline{H}_{T_2} - \Delta\overline{H}_{T_1} = \int_{T_1}^{T_2} \Delta\overline{C_P}dT$$

This new equation, relating enthalpy changes for a reaction at different temperatures to the integrated change in specific heat between those temperatures, is known as *Kirchhoff's Law*. It's typically simplified two ways. The first is assuming T_1 is the reference enthalpy change, at standard ambient temperature and pressure. We can therefore rewrite $\Delta\overline{H}_{T_1}$ as $\Delta\overline{H}^{\circ}$, assuming that the standard enthalpy change is always associated with a value of T_1 = 298 K. (T_2 becomes T_{new} in this arrangement.) The second is the usually-satisfactory assumption that constant-pressure specific heats don't change substantially with temperature - indeed, many of the equations we've derived previously assume that $\overline{C_P}$ values don't change with temperature. This brings us to the form of Kirchoff's Law we will normally use:

$$\Delta\overline{H}_{T_{new}} = \Delta\overline{H}^{\circ} + \Delta\overline{C_P}(T_{new} - 298\text{K})$$

This brings us to a new, practical relationship. If we know the standard enthalpy change for a reaction and we know how the specific heats of the substances in that reaction change, we can know the enthalpy change for that reaction at any temperature T_{new}.

CHAPTER OVERVIEW

3: Heat engines and the second law

We all know of engines. We could talk about them in general terms - the thing that runs our car, and such as that. If we're really literate, we can talk about exactly how a piston takes gasoline and a spark and generates a great deal of power that keeps us moving forward.

But in most of the contexts we think about engines in our modern life, we don't *really* want to think of engines. We want engines to work, and that's all; they serve the role of black boxes under the hoods of our cars, functioning in ways that we don't care about to get us from place X to place Y as fast as possible.

Maybe I shouldn't speak for you. Maybe you really care about the piston and the cam, all the time. But I know I don't.

So starting out a new section of this course with a definition of an *engine* is awkward - at least for me. But it's necessary because of the process energies we've been deliberately developing over the last section - heat and work. *An engine is a system that transforms heat energy into work performed by the system.*

We care about engines because they're practical applications of many of the ideas we've been talking about in the course thus far. They're the kinds of systems that bring many of our core ideas to bear.

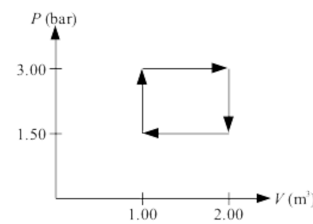
If we're not careful, though, we won't bring those ideas to bear *usefully*. As we get used to the kinds of processes we need to study engines, we can impose conditions on the behavior of them that are unrealistic - and not merely unrealistic in the real world, but unrealistic even in the context of the kind of theoretical study we're trying to build.

So our primary goal, even theoretically, is to think about the engine as practically as possible. We want to be aware of the very real problems that come with engine operation, and we want to solve those problems - or, at least, consider those problems - as completely as we can. If we do this seriously enough, we will come to some conclusions about how well we can transform heat energy into practical work - and those conclusions will complete our picture of thermodynamics.

3.1: A first heat engine - the PV rectangle

Let's solve a relatively simple problem first.

A monoatomic ideal gas begins with a volume of 2.00 m^3 and under a pressure of 1.50 bar and at a temperature of 298 K. The gas first is compressed to a volume of 1.00 m^3 at constant pressure, then (in the compressed state) pressurized to a pressure of 3.00 bar at constant volume, then decompressed back to 2.00 m^3 at constant pressure, then restored to its original state in every way - the number of moles have not changed, and the temperature is again 298 K. Find the total work done by the gas at each step, and the total heat added during the cycle.



I've provided a sketch of this cycle at the right. We can dub this a *constant pressure / constant volume* cycle, or simply a *PV rectangle*, because the constant pressure and constant volume steps make the cycle appear very simple. The cycle starts at the lower right corner of the rectangle, with the expanded gas at lowest pressure, and then proceeds clockwise around the rectangle.

Because of the integral definition of work ($w = -\int P\, dV$), we could find the total work for the cycle by simply finding the area of the rectangle. However, at this point, it will be more instructive to work the exercise out, step by step.

In step 1, the *compression step*, we take a gas at 2.00 m^3 and, while holding it at a constant volume of 1.50 bar (which equals 1.50 × 10^5 Pa), we halve its volume. This is the arrow at the bottom of the rectangle on the graph. It is very straightforward to find the work done on this gas (which is positive, because volume is decreasing):

$$w_1 =-P_1\,\Delta_1\,V =-(1.50\times 10^5\mathrm{Pa})(1.00\mathrm{m}^3 - 2.00\mathrm{m}^3) = 1.50\times 10^5\,\mathrm{J}(\text{compression})$$

(There is new notation here that you are probably not accustomed to. The subscript is applied to the delta, writing $\Delta_1 V$ and not ΔV_1, to emphasize that this is the first *change* in volume, and not a change in "volume 1". This first change in volume is the compression between our two given volumes; we'll distinguish this from the expansion to come.)

In the second step (represented by the arrow at the left of the graph), where the gas is *pressurized*, it is held at constant volume. *If volume of a gas remains constant, no work is done.* $w_2 = 0$ for the second step.

In step 3, the *expansion step*, we now have a gas at a heightened pressure (3.00 bar, or 3.00 × 10^5 Pa), and we are letting the volume of the gas relax back to 2.00 m^3. This work is also easily computed:

$$w_3 =-P_3\,\Delta_3\,V =-(3.00\times 10^5\mathrm{Pa})(2.00\mathrm{m}^3 - 1.00\mathrm{m}^3) = -3.00\times 10^5\,\mathrm{J}(\text{expansion})$$

And again, in the fourth step, where the gas is *depressurized*, it is held at constant volume. *If volume of a gas remains constant, no work is done.* $w_4 = 0$ for the fourth step.

Let's summarize the calculations we've made:

$$w_1 = 1.50\times 10^5\,\mathrm{J}(\text{compression})$$

$$w_2 = 0(\text{pressurization})$$

$$w_3 = -3.00\times 10^5\,\mathrm{J}(\text{expansion})$$

$$w_4 = 0(\text{depressurization})$$

$$w_1 + w_2 + w_3 + w_4 = 1.50\times 10^5\,\mathrm{J} + 0 - 3.00\times 10^5\,\mathrm{J} + 0 = \mathbf{-1.50\times 10^5 J = -150kJ}$$

Note a couple of details in this work. We have completed a full cycle of the compression, pressurization, decompression, and depressurization. The total work done has a value; it is not zero. (Given the integration we talked about earlier, find the area of the rectangle and see if you understand where the total work done came from.)

What's more, if the difference in the pressures is any greater, the value of this work increases in kind, without any kind of limit. This wouldn't be offensive at all if it wasn't for the entirely unrealistic constraint in the problem - no moles of gas lost, and the temperature returns to exactly the same value it started from.

This would require that, because the initial and final states are the same,

$$\Delta U=\frac{3}{2}nR\Delta T=\frac{3}{2}nR(0)=0(\text{given}\Delta T=0\text{for full cycle})$$

Because ΔU = 0, by the First Law:

$$\Delta U=q+w\rightarrow\text{if}\Delta U=0,q=-w$$

Therefore, if the change in internal energy is zero, *we don't have to do any further computation to find the total heat transferred.* If the work done by the system is –150 kJ, then **the heat transferred into the system must be q = 150 kJ.**

Our hope, therefore, is that when we compute the heat transfer for each step, we come back to the same result. To do this, we must now concern ourselves with the conditions at each step.

Initially we're at 2.00 m^3, 1.50 bar (which is equal to 1.50 × 10^5 Pa) and 298 K. We need a number of moles (note the value of R we use):

$$n=\frac{PV}{RT}=\frac{(1.50\times10^5\text{Pa})(2.00\text{m}^3)}{(8.3145\text{J mol}^{-1}\text{K}^{-1})(298\text{K})}=12\underline{1}.1\text{mol}$$

This allows us to compute the temperatures at *each successive* set of conditions, and prepare for the temperature to fluctuate more than a little bit:

$$\text{Compressed:}T=\frac{PV}{nR}=\frac{(1.50\times10^5\text{Pa})(1.00\text{m}^3)}{(121.1\text{mol})(8.3145\text{J mol}^{-1}\text{K}^{-1})}=149\text{K}$$

$$\text{Pressurized:}T=\frac{PV}{nR}=\frac{(3.00\times10^5\text{Pa})(1.00\text{m}^3)}{(121.1\text{mol})(8.3145\text{J mol}^{-1}\text{K}^{-1})}=298\text{K}$$

$$\text{Decompressed:}T=\frac{PV}{nR}=\frac{(3.00\times10^5\text{Pa})(2.00\text{m}^3)}{(121.1\text{mol})(8.3145\text{J mol}^{-1}\text{K}^{-1})}=596\text{K}$$

$$\text{Depressurized: (you can demonstrate that the temperature returns to 298 K!)}$$

Knowing temperatures, we can compute heat transfers at each step. Let's be very careful about how we separate the quantities we're studying, because we've set ourselves up to solve a problem like this.

Two steps are steps of changing volume - the *compression* and the *decompression*. During these steps, the pressure of the container is held constant. And, because this is a monoatomic ideal gas, we know what the molar specific heat is - it's $^5/_2R$. The other two steps are the *pressurization* and *depressurization* - during those steps, it's volume held constant, and the molar specific heat becomes $^3/_2R$.

Let's build out each of these computations:

$$q_1=n\overline{C_P}\Delta T=(12\underline{1}.1\text{mol})\frac{5}{2}(8.3145\text{J mol}^{-1}\text{K}^{-1})(149\text{K}-298\text{K})=-37\underline{5}.1\text{kJ}(\text{compression})$$

$$q_2=n\overline{C_V}\Delta T=(12\underline{1}.1\text{mol})\frac{3}{2}(8.3145\text{J mol}^{-1}\text{K}^{-1})(298\text{K}-149\text{K})=22\underline{5}.0\text{kJ}(\text{pressurization})$$

$$q_3=n\overline{C_P}\Delta T=(12\underline{1}.1\text{mol})\frac{5}{2}(8.3145\text{J mol}^{-1}\text{K}^{-1})(596\text{K}-298\text{K})=75\underline{0}.3\text{kJ}(\text{decompression})$$

$$q_4=n\overline{C_V}\Delta T=(12\underline{1}.1\text{mol})\frac{3}{2}(8.3145\text{J mol}^{-1}\text{K}^{-1})(298\text{K}-596\text{K})=-45\underline{0}.0\text{kJ}(\text{depressurization})$$

$$q_1+q_2+q_3+q_4=-37\underline{5}.1\text{kJ}+22\underline{5}.0\text{kJ}+75\underline{0}.3\text{kJ}-45\underline{0}.0\text{kJ}=\mathbf{150kJ}$$

It's always reassuring when a long computation works correctly!

There's one other detail hidden behind these heat computations. The entire point of a heat engine is to take heat in to do useful work. We have the work done by the engine: -150 kJ. The heat we took *in*, though, is much, much greater - it's the sum of the two positive values of q, those associated with the pressurization and decompression steps, when the temperature of the engine was increasing. That's 975 kJ.

Of those 975 kJ of energy, we only used 150 kJ of it. The rest of the energy - 825 kJ - was simply kicked out of the engine, as waste heat.

We can define a quantity for an engine, the *efficiency*, that describes the fraction of the the heat energy input that we use to do useful work. Because of the sign on work when the engine does work on the surroundings, we have to have an extra negative sign:

$$e = -\frac{w}{q_{ABS}} = -\frac{-150\mathrm{kJ}}{975\mathrm{kJ}} = 0.154 (= 15.4\%)$$

An engine that only uses 15.4% of absorbed heat to do useful work is pretty darn inefficient. We need to do better. We need to perform using cycles that go beyond simple constant-pressure and constant-volume steps.

3.2: Steps to improve efficiency: isothermal and adiabatic changes

There are two special cases we need to consider in order to extend the way we construct these cycles beyond the constant volume (or *isochoric*) and the constant-pressure (or *isobaric*) circumstances.

One case concerns a process in which there's no temperature change, but where work is done. We've said before that in most processes where there's no temperature change involve no heat transfer - but we've also said that temperature most closely connects to the state of internal energy. When those two statements come into conflict, the state function "wins out" - in an *isothermal* process where work is done, there will be heat transfer; internal energy change is forced to zero.

To put it another way, if the temperature of a system (and therefore that system's internal energy) is held constant, any heat you introduce to the system *has* to immediately be released from the system as work, and any work you do on the system *has* to immediately released from that system as heat.

We have, though, already found what work equals in a state of constant temperature:

$$w = nRT \ln \frac{V_i}{V_f}$$

If $\Delta U = 0$, and $q = -w$, then this must also be true (note the swapping of subscripts):

$$q = nRT \ln \frac{V_f}{V_i}$$

This expression is referred to as the *isothermal transfer of heat*. Even though the pressure is changing, we only need the change in volume to complete the computation. And what's more, it aids in generating efficiency; it is a step that encourages total transfer of heat energy to work performed.

We've also studied another way to improve efficiency: preventing heat losses at all. Recall that processes that proceed in a fashion that prevent heat loss are known as *adiabatic* processes. These are either processes that are instantaneous, so that work done happens so rapidly that heat doesn't have time to transfer, or processes that are well-insulated, so temperature changes inside the container are unable to impact temperature outside of the container. In these cases, because $q = 0$, all work done on or by the system translates into the internal energy change of the system; $\Delta U = w$.

We have been through the realization that adiabatic processes don't neatly translate to holding a quantity constant in the same way that constant-volume, constant-pressure, or isothermal processes do. We have to intentionally derive the quantity that's being held constant.

It turns out that we've already laid the foundation for that derivation, when we worked out how volume changes with changing temperature in an adiabatic process:

$$\frac{T_f}{T_i} = \left(\frac{V_i}{V_f}\right)^{\gamma-1} where \gamma = \frac{\overline{C_P}}{\overline{C_V}} (\text{adiabatic process})$$

We can rearrange this to group final and initial terms together on their respective sides of the equation:

$$T_f V_f^{\gamma-1} = T_i V_i^{\gamma-1} (\text{adiabatic process})$$

$$TV^{\gamma-1} = constant (\text{adiabatic process})$$

It turns out (and this is left as an exercise to the reader) that we can derive a very similar equation by working out how pressure changes with changing volume in an adiabatic environment. Here, we recall Boyle's law where temperature is held constant in a closed container:

$$PV = constant (\text{closed container at constant temperature})$$

It turns out that, similar to Boyle's law, pressure does increase when the volume of the container decreases in an adiabatic process. However, in an adiabatic process, temperature is *not* intentionally held constant, and therefore the relationship looks subtly different, and reintroduces γ:

$$P_f V_f^{\gamma} = P_i V_i^{\gamma} (\text{adiabatic process})$$

$$PV^{\gamma} = constant (\text{adiabatic process})$$

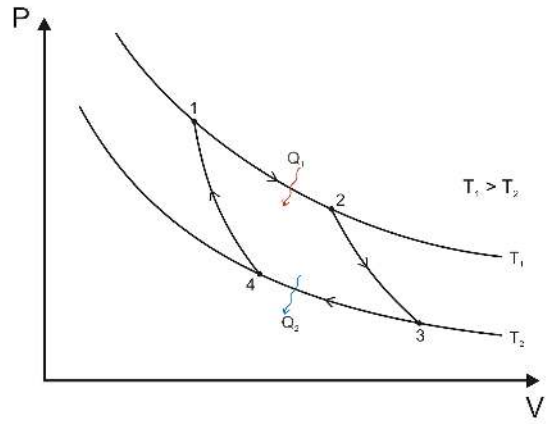
The ratio γ is easily shown to be a precise ratio for ideal gases; it's 5/3 for monatomic ideal gases, and 7/5 for diatomic. (Prove this!) For real gases, the ratio isn't quite as neat, but it is viable to work with.

Being able to equate internal energy change with work done for no heat transfer makes the computation of work done very straightforward even if pressure isn't held constant ($\Delta U = w = n\overline{C_V}\Delta T$). And this gives us the tools to develop a heat engine that is much more efficient than the constant-pressure/constant-volume cycle that we described before.

Here is a table that summarizes the types of processes we have revealed so far, and that we can use as reference for determining internal energy changes, heat transfers, and work done:

Name of process	**Constant quantity**	$\boldsymbol{\Delta} U=$	$q=$	$w=$
Isobaric	P	$n\overline{C_V}\Delta T$	$n\overline{C_P}\Delta T$	$-P\Delta V$
Isochoric	V	$n\overline{C_V}\Delta T$	$n\overline{C_V}\Delta T$	0
Isothermal	T	0	$nRT\ln(V_f/V_i)$	$nRT\ln(V_i/V_f)$
Adiabatic	$TV^{\gamma-1}; PV^{\gamma}$	$n\overline{C_V}\Delta T$	0	$n\overline{C_V}\Delta T$

3.3: The Carnot engine



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A single book was the entire scientific output of the military scientist Sadi Carnot; it was called *Reflections on the Motive Power of Fire*, and as the title implies, it describes the behavior of all kinds of heat engines. Carnot was driven to maximize the efficiency of the engine, and by so doing, he was attracted to the steps that both eliminated heat transfer and concentrated heat input into useful work.

The engine that Carnot devised that focused on these steps started in the upper-left corner of the new diagram at the top of this page, at the highest pressure and volume. The step where heat energy would be invested, in Carnot's vision, would be an *isothermal* step that both increased the pressure on the gas and decreased the volume, it performed large quantities of work at high temperature. The step that followed would perform more work by further reducing pressure and volume without heat transfer, in a rapid *adiabatic* process.

Those steps would then need to be undone, but the adiabatic process would lower the temperature considerably, so the heat released by the engine would be less in the third step and the "wasted work" in the final two steps would be much lower.

Here's an example problem laying out such a cycle, using the Carnot cycle figure:

A monatomic ideal gas begins with a volume of 1.00 m^3 and under a pressure of 3.00 bar and at a temperature of 596 K. The gas first does work by decompressing and depressurizing isothermally at 596 K; the pressure at the end of the isothermal step is 2.00 bar). The gas then decompresses and depressurizes adiabatically to a temperature of 298 K. Work is then done to the gas to compress and pressurize it isothermally at 298 K, and then it is restored to its original state by an adiabatic step; the number of moles have not changed, and the temperature is again 596 K. Find the total work done by the gas at each step, and the total heat added during the cycle.

Note that this problem mirrors the problem given you in the PV rectangle, but for isothermal and adiabatic steps instead - the curves in lieu of the straight horizontal and vertical lines on the graph. We can't just use the ideal gas law all the time.

We *can*, however, use the ideal gas law to find the number of moles of gas present from the initial conditions of the gas:

$$n = \frac{P_1V_1}{RT_1} = \frac{(3.00 \times 10^5 \mathrm{Pa})(1.00 \mathrm{m}^3)}{(8.3145 \mathrm{J\ mol}^{-1} \mathrm{K}^{-1})(596 \mathrm{K})} = 60.\underline{5}4 \mathrm{mol}$$

Note from the problem that V_1 = 1.00 m^3 and that the pressures given are P_1 = 3.00 × 10^5 Pa and P_2 = 2.00 × 10^5 Pa, respectively. From there, knowing that temperature is held constant, we can find the volume at the end of the cycle:

$$V_2 = \frac{nRT_2}{P_2} = \frac{(60.54 \mathrm{mol})(8.3145 \mathrm{J\ mol}^{-1} \mathrm{K}^{-1})(596 \mathrm{K})}{2.00 \times 10^5 \mathrm{Pa}} = 1.5\underline{0} \mathrm{m}^3$$

The other pressures and volumes we need come from the adiabatic relations. Because we are studying a monatomic ideal gas, $\gamma = {}^5/_3$:

$$T_2V_2^{\gamma-1} = T_3V_3^{\gamma-1} \rightarrow V_3 = \left(\frac{T_2V_2^{\gamma-1}}{T_3}\right)^{\frac{1}{\gamma-1}} = \left(\frac{(596\text{K})(1.50\text{m}^3)^{\frac{2}{3}}}{298\text{K}}\right)^{\frac{3}{2}} = 4.2\underline{4}3\text{m}^3$$

$$P_2V_2^{\gamma} = P_3V_3^{\gamma} \rightarrow P_3 = P_2\left(\frac{V_2}{V_3}\right)^{\gamma} = (2.00\times10^5\text{Pa})\left(\frac{1.50\text{m}^3}{4.243\text{m}^3}\right)^{\frac{5}{3}} = 3.5\underline{3}5\times10^4\text{Pa}$$

The volume and pressure at the start of step 4 result from the adiabatic step taking us back to the initial condition:

$$T_1V_1^{\gamma-1} = T_4V_4^{\gamma-1} \rightarrow V_4 = \left(\frac{T_1V_1^{\gamma-1}}{T_4}\right)^{\frac{1}{\gamma-1}} = \left(\frac{(596\text{K})(1.00\text{m}^3)^{\frac{2}{3}}}{298\text{K}}\right)^{\frac{3}{2}} = 2.8\underline{2}8\text{m}^3$$

$$P_1V_1^{\gamma} = P_4V_4^{\gamma} \rightarrow P_4 = P_1\left(\frac{V_1}{V_4}\right)^{\gamma} = (3.00\times10^5\text{Pa})\left(\frac{1.00\text{m}^3}{2.828\text{m}^3}\right)^{\frac{5}{3}} = 5.3\underline{0}3\times10^4\text{Pa}$$

This gives us all the data we need to work out the four steps of the cycle. Here, unlike the PV rectangle, we will compute work for all four steps, both decompressions and both compressions; on the other hand, we will only need to compute heat for two of the four:

$$\text{Isothermal dec.:} w_{12} = nRT_1\ln\left(\frac{V_1}{V_2}\right) = (60.54\text{mol})(8.3145\text{J mol}^{-1}\text{K}^{-1})(596\text{K})\ln\left(\frac{1.00\text{m}^3}{1.50\text{m}^3}\right) = -121.6\text{kJ}$$

$$\text{Adiabatic decompression:} w_{23} = n\overline{C_V}\Delta_{23}T = (60.54\text{mol})\frac{3}{2}(8.3145\text{J mol}^{-1}\text{K}^{-1})(298\text{K}-596\text{K}) = -225.0\text{kJ}$$

$$\text{Isothermal comp.:} w_{34} = nRT_3\ln\left(\frac{V_3}{V_4}\right) = (60.54\text{mol})(8.3145\text{J mol}^{-1}\text{K}^{-1})(298\text{K})\ln\left(\frac{4.243\text{m}^3}{2.828\text{m}^3}\right) = 60.86\text{kJ}$$

$$\text{Adiabatic compression:} w_{41} = n\overline{C_V}\Delta_{41}T = (60.54\text{mol})\frac{3}{2}(8.3145\text{J mol}^{-1}\text{K}^{-1})(596\text{K}-298\text{K}) = 225.0\text{kJ}$$

Note that, because they're so closely equated to the state energy change ΔU, the work done in steps 23 and 41 undo one another. This will always be what we expect for an adiabatic step, because heat transferred is zero for those steps.

The total work done is therefore:

$$w_{TOT} = w_{12} + w_{23} + w_{34} + w_{41} = -12\underline{1}.6\text{kJ} - 22\underline{5}.0\text{kJ} + 60.\underline{8}6\text{kJ} + 22\underline{5}.0\text{kJ} = -6\underline{0}.8\text{kJ}$$

(It's just as likely, if you round, that you come up with 6$\underline{0}$.7 kJ instead of 6$\underline{0}$.8 kJ. That's OK. The significant figure rules dictate that the result of this addition and subtraction is only significant to the ones place, which will be what we pay attention when we get to the punch line of this calculation.)

The heat transferred in the isothermal steps will look incredibly similar, whereas the adiabatic steps simplify by definition:

$$\text{Isothermal decomp.:} q_{12} = nRT_1\ln\left(\frac{V_2}{V_1}\right) = (60.54\text{mol})(8.3145\text{J mol}^{-1}\text{K}^{-1})(596\text{K})\ln\left(\frac{1.50\text{m}^3}{1.00\text{m}^3}\right) = 121.6\text{kJ}$$

$$\text{Adiabatic decompression:} q_{23} = 0 \text{by definition}$$

$$\text{Isothermal comp.:} q_{34} = nRT_3\ln\left(\frac{V_4}{V_3}\right) = (60.54\text{mol})(8.3145\text{J mol}^{-1}\text{K}^{-1})(298\text{K})\ln\left(\frac{2.828\text{m}^3}{4.243\text{m}^3}\right) = -60.86\text{kJ}$$

$$\text{Adiabatic compression:} q_{41} = 0 \text{by definition}$$

It should be a very straightforward exercise to demonstrate that the total heat input to the engine is the same magnitude as the total work - **60.8 kJ.**

The efficiency of this engine is calculated in the same way as before, with the heat transferred from the isothermal decompression being the absorbed quantity:

$$e = -\frac{w}{q_{ABS}} = -\frac{-6\underline{0}.8\text{kJ}}{12\underline{1}.6\text{kJ}} = 0.5\underline{0}0 (= 50\%)$$

The efficiency is 50%. It's 50% when the low-temperature isothermal step is half the temperature of the high.

Huh.

3.4: The thermodynamic existence of entropy

There is a fact that's hidden in the solution to our Carnot cycle in section 3.3, and we need a formal derivation in order to dig that fact out.

Let's return to the definition of efficiency:

$$e=-\frac{w}{q_{ABS}}$$

We already have made statements about what work and heat should be for this engine in general; because we're studying a closed cycle for which $\Delta U = 0$, $q = -w$ for the full cycle. Let's rewrite the definition of efficiency specific to the full cycle:

$$e=\frac{q_{OUT}+q_{IN}}{q_{IN}}$$

The heat input in this cycle is the heat transfer from stage 12, which was positive. The heat output from this cycle was the heat transfer from stage 34, which was negative:

$$q_{IN}=nRT_1\ln\left(\frac{V_2}{V_1}\right)$$

$$q_{OUT}=nRT_3\ln\left(\frac{V_4}{V_3}\right)$$

If we substitute these expressions into the efficiency definitions, we'll be able to cancel out n and R - but the natural logarithm expressions won't cancel. But the adiabatic steps in the Carnot cycle connect temperature and volume, and there are only two temperatures - the high temperature T_1, and the low temperature T_3. (The temperature at state 1, after all, is isothermal to the temperature at state 2; likewise the temperatures at state 3 and state 4.) We can relate the two volume ratios using the adiabatic relations…

$$T_1V_2^{\gamma-1}=T_3V_3^{\gamma-1}\rightarrow\frac{T_3}{T_1}=\frac{V_2^{\gamma-1}}{V_3^{\gamma-1}}$$

$$T_1V_1^{\gamma-1}=T_3V_4^{\gamma-1}\rightarrow\frac{T_3}{T_1}=\frac{V_1^{\gamma-1}}{V_4^{\gamma-1}}$$

We can set both right sides equal to one another, eliminate the exponents, and rearrange for equality's sake…

$$\frac{V_2^{\gamma-1}}{V_3^{\gamma-1}}=\frac{V_1^{\gamma-1}}{V_4^{\gamma-1}}\rightarrow\frac{V_2}{V_1}=\frac{V_3}{V_4}$$

This would have been *perfectly convenient* if $V_2 / V_1 = V_4 / V_3$. As it stands, though, we're still okay because we can rearrange q_{IN} with a negative log:

$$q_{IN}=nRT_1\ln\left(\frac{V_2}{V_1}\right)=-nRT_1\ln\left(\frac{V_4}{V_3}\right)$$

Now we substitute q_{IN} and q_{OUT} into the efficiency definition and simplify:

$$e=\frac{q_{OUT}+q_{IN}}{q_{IN}}=\frac{nRT_3\ln(V_4/V_3)-nRT_1\ln(V_4/V_3)}{-nRT_1\ln(V_4/V_3)}=\frac{T_3-T_1}{-T_1}$$

Here we have our demonstration that the Carnot engine's efficiency only depends on temperature. Again, remember that there are only two temperatures - the "HOT" temperature T_1, and the "COLD" temperature T_3. Further simplification gets us a classic expression:

$$e=-\frac{T_3}{T_1}+\frac{T_1}{T_1}=1-\frac{T_3}{T_1}$$

$$e = 1 - \frac{T_{COLD}}{T_{HOT}}$$

That expression is the ultimate statement of the futility of chasing a perfect engine. T_3/T_1 or T_{COLD}/T_{HOT} must be zero if the efficiency of the engine is to be a perfect 1, and all the input heat is to be transformed into useful work. There are only two ways we can get that ratio to zero; either the hot-temperature reservoir must have an infinite temperature (lol), or the cold-temperature reservoir must be absolute zero (which is physically impossible).

There are no perfect engines. Every engine that you could ever create would contribute heat energy to the universe.

This, ultimately, is the thermodynamic significance of entropy; you can extend this thinking to the understanding that *every transfer of energy* adds heat energy, unusable thermal chaos, to the universe.

3.5: Entropy defined

The work we did to get to this description of the efficiency of the Carnot engine has a side benefit. The fact that we could cancel everything about the heat expressions but temperature means that the heat transferred at the two isothermal steps of the Carnot engine's function is proportional to the temperature at those steps (with a negative sign embedded because of the sign flip we did in the final substitution). We can build out an expression to describe this better:

$$\frac{q_3}{T_3} = -\frac{q_1}{T_1} \text{OR} \frac{q_{\text{OUT}}}{T_{\text{COLD}}} = -\frac{q_{\text{IN}}}{T_{\text{HOT}}}$$

The ratio of heat to temperature will be a useful expression going forward; in real engines, the equivalency will not be neat, and the magnitude of the ratio of heat out to the cold temperature *will always* be greater than the magnitude of the ratio of the heat in to high temperature, because in real engines heat will always be lost in the energy transformation. This ratio of heat to temperature will be how we formalize the definition of *entropy change* at a single temperature:

$$\Delta S = \frac{q}{T}$$

A positive entropy change always corresponds with heat being absorbed by a system, and the disorder in that system increasing. A negative entropy change always corresponds with heat being released by that system, and the disorder in that system decreasing.

One of the things that becomes clear when going over a multiple of sources for studying physical chemistry (even when comparing these notes to textbooks I've adopted for physical chemistry in the past) is that there is a spectrum of approaches to introducing and discussing entropy. I have obviously focused on engines, and the disorder that must be generated by the proper function of an engine. A introductory physical chemistry text will typically focus on the phases of matter; a gas phase has more entropy than a liquid phase, which has more entropy than a solid phase. (There is an absolute definition of entropy, which is statistical; we will spend very little time on that definition. We will far more often talk of entropy change than we will of absolute entropy.)

However, one thing that will become plain as we move forward in our dealings with entropy is that entropy does *not* always change at a single temperature, and we will not be able to make the assumption that all processes where entropy changes are isothermal. We need to work with differential language:

$$dS = \frac{\delta q}{T}$$

Remember that we write differential heat as δq and not as dq because heat is a process variable, and we can't integrate over heat unless we integrate the entire process. If we are going to engage in a differential process, and we're going to be able to identify the limits of that integration, we need to be able to transform differential heat into something that we can properly integrate over.

We know, for example, what heat transfer looks like at constant pressure:

$$q_P = n\bar{C_P}\Delta T$$

If temperature change is only as large as a differential, though, we can express that heat transfer by means of a differential that will be integrable while δq is not:

$$\delta q_P = n\bar{C_P}dT$$

This is an expression we can integrate and find a change in entropy for!

$$dS = \frac{\delta q_P}{T} = n\bar{C_P}\frac{dT}{T}$$

$$\int_{S_i}^{S_f} dS = n\bar{C_P}\int_{T_i}^{T_f}\frac{dT}{T}$$

$$\Delta S = n\bar{C_P}\ln\left(\frac{T_f}{T_i}\right)$$

We find the entropy change for matter that's changing its temperature through heat transfer by taking the natural logarithm of the ratio of change of temperatures. Note that while this is temperature-dependent, the logarithm is unitless; the units of entropy change are energy units per unit temperature, or J K^{-1} in base SI units.

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